

Department of Aeronautics

MSc Programmes: Project Topics 2022-23

MSc in Advanced Aeronautical Engineering

MSc in Advanced Computational Methods for Aeronautics,
Flow Management and Fluid-Structure Interaction

MSc in Composites: The Science, Technology and
Engineering Application of Advanced Composites

**Please note that you are only permitted to select
projects offered to your MSc programme.**

**Please also refer to *MSc Projects Information for
Students and Supervisors 2022-23* for additional
guidance on the project process.**

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Aliabadi, Ferri Prof

Project no:	IC-COMFA01
Project title:	Critical Assessment of Damage Detection Methodologies for Structural Health Monitoring in Composite Structures
Supervisor:	Aliabadi, Ferri
Category:	Theoretical;#Critical Review
Software:	
Confidential:	Yes
Offered to:	
	MSc in Composites

Structural health monitoring (SHM) is a relatively new and alternative way of non-destructive inspection (NDI). It is the process of implementing a damage detection and characterization strategy for composite structures. The basis of SHM is the application of permanent fixed sensors on a structure, combined with minimum manual intervention to monitor its structural integrity. These sensors detect changes to the material and/or geometric properties of a structural system, including changes to the boundary conditions and system connectivity which adversely affect the system's performance.

Ultrasonic guided wave has been recognised as an effective tool for interrogating large structures due to its capability of propagating over long distances with small energy loss. Typically, a guided wave structural health monitoring (GWSHM) system utilises a network of piezoelectric transducers to excite guided wave and capture the guided wave signals. Detection of damage relies on interpreting the correlation between change of the signals and the characteristics of damage. In aircraft structures, where structural complexities such as multiple boundaries, stiffeners and openings are typical, guided wave signals inevitably contain multiple boundary scatters and reflections, which makes the captured signal difficult to interpret. In this case, pre-recorded measurements in a known integrity state of the structure serve as baselines for identifying deviations from the baseline measurements and can be related to the occurrence of damage.

Projects require one or more other following skills: Knowledge of Structural Analysis

Project no:	IC-MSCFA02
Project title:	Critical Assessment of Early Warning Impact Detection using Structural Health Monitoring
Supervisor:	Aliabadi, Ferri
Category:	Theoretical;#Critical Review
Software:	
Confidential:	
Offered to:	MSc in Advanced Aeronautical Engineering
	MSc in Composites

Composite materials are prone to barely visual impact damage which can cause significant loss in strength but is difficult to detect without detailed inspection. In the aviation industry, real-time characterization of impact events has a direct influence on design, production and maintenance costs. Impact characterization consists of estimating the location and magnitude of the impact by using the data collected via sensors. This is an inverse problem; the input to the system is estimated from its response. In practice, the response of the structure subjected to an impact is recorded via embedded or surface-mounted sensors. These data are subsequently analysed with a suitable algorithm to locate the impact and determine the peak impact force or possibly reconstruct the impact force history. Therefore, there is growing interest to develop a reliable sensor based Structural Health Monitoring (SHM) system for impact detection in composite structures. The availability of such system may reduce inspection requirements and design damage tolerance requirements leading to cost savings. For continuous monitoring, passive sensing is preferred due to no need for continuous active actuation for monitoring. Passive sensing methods for impact localisation can be generalised into 2 groups: analytical and data driven. Analytical methods such as triangulation depend on knowledge of the structural behaviour (such as wave propagation speed) to predict the impact location based on extracted features from the sensor recorded structural responses (e.g. impact wave Time of Arrival [TOA] to sensor). Data driven methods also utilise extracted features from sensor recordings but do not require specific knowledge of the structural behaviour to deduce the impact location.

Instead, data driven methods use previously collected training data from the structure as a reference for deducing impact locations.

Project no:	IC-ACMFA03
Project title:	Boundary Element Analysis of Crack Growth
Supervisor:	Aliabadi, Ferri
Category:	Computational
Software:	In-house code, Fortran
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Catastrophic fracture failure of engineering structures is caused by cracks that extend beyond a safe size. In this project crack growth processes are simulated with the boundary element method. Boundary element method is an alternative method to the more established finite element method. The attraction of the boundary element method is largely attributed to the reduction in the dimensionality of the problem; for two-dimensional problems, only the line-boundary of the domain needs to be discretized into elements, and for three-dimensional problems only the surface of the problem needs to be discretized. This means that, compared to the FEM, a boundary element analysis results in a substantial reduction in modelling effort. Boundary element has been particularly successful for crack growth analysis. The aim of the project is to extend the current in-house code to include thermal, centrifugal and other types of body forces.

Prerequisite: Programming, good mathematical skills

Project no:	IC-COMFA04/E
Project title:	Environmental effects on damage detection in composite panels using Structural Health Monitoring techniques
Supervisor:	Aliabadi, Ferri
Category:	Experimental;#Computational
Software:	Experimental/ Matlab/LabView
Confidential:	No
Offered to:	
	MSc in Composites

Structural Health Monitoring (SHM) techniques are well established for damage detection at laboratory conditions where the environmental factors are controlled. However, applying the SHM methodologies to real aircraft structure under real load conditions and environmental effects might not result in correct diagnosis of the structure. Therefore it is very important to study the effect of variable such as temperature and humidity on the wave propagation in composite structures. Most of the damage detection techniques based on guided waves are developed on the basis of comparing the response of the structure in its pristine state to the damaged state. This means that it is very important to study the influence of the environment on the sensor data when they are recorded under different conditions to avoid miss-detections or false alarms due to environmental factors. In addition pre-loading of the structures can also change the propagational properties of the baseline signals. Therefore the aim of the project is to identify the influence of the environmental effects together with

the impact of the pre-loading on the wave propagation and possibly develop baseline free methods for damage detection in composite aircraft panels.

Prerequisite: Programming with Matlab, background in Structures

Amato, Davide Dr

Project no:	IC-MSCDA01
Project title:	Highly efficient numerical orbit propagation with Taylor methods
Supervisor:	Amato, Davide
Category:	Computational;#Numerical
Software:	heyoka (Python); Python, C++
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;#
	MSc in Advanced Computational Methods

Numerical orbit propagation consists in the solution of the equations of motion for a spacecraft under the gravitational acceleration of a massive body (i.e., the Earth) and additional perturbations. This is an extremely important task, which is computationally intensive especially when high-fidelity perturbations and many spacecraft are considered. Taylor integration solves an Initial Value Problem (IVP) for ODEs by expanding the dynamics in a high-order Taylor series. Recently, a highly efficient Taylor integration method has been proposed exploiting automatic differentiation (AD) to perform very fast integrations in astrodynamics. The method, which is implemented in the heyoka Python/C++ library [1], has shown excellent performance in the integration of trajectories in the Solar System, outperforming state-of-the-art numerical integration methods.

The primary goal of the project is to enhance the numerical efficiency of heyoka by implementing regularised equations of motion, instead of the classical Cowell method. Regularisation consists in removing singularities from the equations of motion for a spacecraft through analytical manipulations. The independent variable is changed from physical time to a “fictitious time”, which is an angle-like quantity, through a differential transformation called the Sundman transformation. The core advantage of regularisation is that the resulting non-singular equations of motion are “smoother”, that is to say, their higher derivatives are smaller. The resulting equations are easier to integrate numerically and exhibit better numerical stability properties.

The secondary goal of the project consists in adapting the method to the integration of the perturbed two-body problem for Earth satellite orbits, and evaluating whether the numerical efficiency is preserved when implementing complex perturbations such as a high-degree geopotential and atmospheric drag.

The project requires an excellent mathematical background, willingness to go beyond the course curriculum, and knowledge of orbital dynamics. Familiarity is required with Python and desired with C++.

[1] F. Biscani and D. Izzo, ‘Revisiting high-order Taylor methods for astrodynamics and celestial mechanics’, Monthly Notices of the Royal Astronomical Society, vol. 504, no. 2, pp. 2614–2628, Jun. 2021, doi: 10.1093/mnras/stab1032.

Project no:	IC-MSDA02
Project title:	Regularised orbit propagation in the circular, restricted three-body problem
Supervisor:	Amato, Davide
Category:	Theoretical;#Numerical
Software:	MATLAB; Julia
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

The circular, restricted three-body problem (CR3BP) consists in solving the motion of a massless object affected by the gravitational accelerations of two massive bodies orbiting each other on circular orbits. The CR3BP is an extremely important problem in celestial mechanics and astrodynamics, as it is a good approximation of many physical systems, such as spacecraft travelling in the Earth-Moon system. This is particularly relevant in light of the renewed interest in the exploration of the Moon driven by the NASA Artemis programme and cislunar Space Domain Awareness. The CR3BP has a rich dynamical behaviour, and many of the periodic orbits in the system are unstable, thus hard to simulate numerically. Regularisation consists in removing singularities from the equations of motion for a spacecraft through analytical manipulations. The independent variable is changed from physical time to a “fictitious time”, which is an angle-like quantity, through a differential transformation called the Sundman transformation. The core advantage of regularisation is that the resulting non-singular equations of motion are “smoother”, that is to say, their higher derivatives are smaller; in fact, it is possible to fully linearise the equations of motion for the two-body problem and for the planar CR3BP [1]. The resulting equations are easier to integrate numerically and exhibit better numerical stability properties.

The project will focus on the software implementation of one or more regularisations of the CR3BP, and on the comparison of their numerical efficiency compared to the integration of Cartesian coordinates in either the inertial or synodic reference frames. We will also explore the derivation of a perturbation method to consider additional perturbations in the CR3BP, such as those from additional massive bodies and radiation pressure.

The project requires an excellent mathematical background, willingness to go beyond the course curriculum, and knowledge of orbital dynamics.

[1] V. Szebehely, Theory of Orbits: the restricted problem of three bodies. New York, NY, USA: Academic Press Inc., 1967.

Project no:	IC-MSDA03
Project title:	Rapid generation of entry, descent, and landing trajectories - Project 1
Supervisor:	Amato, Davide
Category:	Computational;#Theoretical
Software:	MATLAB
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Robust Entry, Descent, and Landing (EDL) guidance algorithms are key technologies to enable full launch vehicle reusability and future crewed exploration missions to the Moon

and Mars. EDL guidance is particularly challenging, as EDL dynamics are strongly non-linear and often uncertain.

For preliminary mission design and testing of EDL guidance algorithms, a first guess for the initial conditions at the entry interface (EI) or at powered descent initiation (PDI) is required. In addition, fixed-time algorithms also require an initial guess for the final time. All of these conditions are then subsequently optimised, and the EDL guidance algorithm steers the vehicle towards the target while respecting constraints.

Two projects are available, both of which will be based on fixed-time polynomial guidance algorithms and will be developed in collaboration with Airbus UK [1]:

- 1) Establishing a method for the rapid generation of initial conditions and time-of-flight for EDL trajectories. This will be accomplished through shooting methods (either single- or multiple-shooting), or through global optimisation methods such as genetic algorithms.
- 2) Creating a six degree of freedom simulation framework for testing polynomial guidance algorithms under reasonably realistic vehicle constraints, such as those on heat and mechanical loads.

We will consider test cases involving the EDL trajectories of launcher first stages on the Earth and of future entry vehicles at Mars.

Both projects require a considerable amount of effort during the year. A strong mathematical background is needed for this project, and knowledge of space and entry flight mechanics is a plus.

[1] P. Lu, 'Theory of Fractional-Polynomial Powered Descent Guidance', Journal of Guidance, Control, and Dynamics, vol. 43, no. 3, pp. 398–409, 2020, doi: 10.2514/1.G004556.

[2] N. X. Vinh, A. Busemann, and R. D. Culp, Hypersonic and Planetary Entry Flight Mechanics. Ann Arbor, MI, USA: The University of Michigan Press, 1980.

Project no:	IC-MSCDA04
Project title:	Rapid generation of entry, descent, and landing trajectories - Project 2
Supervisor:	Amato, Davide
Category:	Computational;#Theoretical
Software:	MATLAB
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

See description above.

Project no:	IC-MSFDA05
Project title:	Data-driven space system identification - Project 1
Supervisor:	Amato, Davide
Category:	Computational;#Theoretical
Software:	Chebfun (MATLAB), Keras (Python), Tensorflow (Python)
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Satellites orbiting the Earth are subject to a variety of accelerations in addition to main gravity. These may be due to the non-sphericity of the Earth, interactions with atmosphere, thrust, etc. Modelling these perturbing accelerations is complex, and often requires the knowledge of quantities that are unknown or very hard to model. This is a crucial task to achieve accuracy for long-term trajectory predictions.

The projects will involve modelling of perturbing accelerations from observational data of space objects. This is framed as a “partial” system identification problem, in which part of the dynamics is unknown and must be reconstructed from data. We will examine the potential of different methods such as polynomial and trigonometric regression, and regression using recurrent neural networks.

Both projects are very ambitious and require a considerable amount of effort during the year. A strong mathematical background is needed for this project, and knowledge of space and entry flight mechanics is a plus.

Project no:	IC-MSFDA06
Project title:	Data-driven space system identification - Project 2
Supervisor:	Amato, Davide
Category:	Computational;#Theoretical
Software:	Chebfun (MATLAB), Keras (Python), Tensorflow (Python)
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

See project description above.

Project no:	IC-MSFDA07
Project title:	Time frequency analysis of Earth satellite orbits
Supervisor:	Amato, Davide
Category:	Theoretical;#Numerical
Software:	MATLAB
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Frequency analysis methods offer deep insights into the dynamics of objects in near-Earth space. By exploiting the frequency spectrum of the trajectory expressed in an appropriate state vector form it is possible to easily perform averaging and understand long-term trends. This has important implications, for instance to associate fragments of a collision to their parent bodies. However, classical frequency analysis tools are difficult to apply when the frequency content of the trajectory is time-varying, which is the case for most Earth satellite orbits. For instance, Low Earth Orbits are affected by atmospheric drag which causes the mean motion (and thus the frequency content) to slowly increase with time.

We will explore several time frequency analysis methods such as short-time Fourier transforms and wavelet transforms. These will first be applied to suitable toy problems (e.g., to non-linear harmonic oscillators), and then will be particularised to problems in orbital dynamics. The project will involve one or more of the following aspects:

- 1) implementing novel osculating-to-mean trajectory transformations using time frequency methods
 - 2) qualitative characterisation of the two-body problem in the frequency domain according to different perturbations (geopotential, aerodynamics, third bodies...)
 - 3) investigation of strongly non-linear and/or chaotic dynamics in the frequency domain.
- The project requires an excellent mathematical background, willingness to go beyond the course curriculum, and knowledge of orbital dynamics.

Project no:	IC-MSCDA08
Project title:	High-accuracy neural network ODE solutions
Supervisor:	Amato, Davide
Category:	Theoretical;#Computational
Software:	DeepXDE (Python), Tensorflow/PyTorch (Python)
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Considerable research efforts have been devoted recently to using neural networks as surrogate solutions to Partial Differential Equations (PDE), with Physics-Informed Neural Networks (PINNs) emerging as one of the most popular methods [1]. PINNs are feedforward neural networks that are trained to represent the solution to a PDE. This is accomplished by using the PDE residual in the loss function used for training, and finding the parameters of the network that minimize the loss.

This project will focus on using PINNs for the solution of Initial and Boundary Value Problems (IVPs/BVPs) described by Ordinary Differential Equations (ODEs). Efficient algorithms for solving IVPs and BVPs exist since decades, and PINNs have yet to achieve their full potential in terms of computational efficiency and generalization with respect to classical time-marching algorithms. Recently, it has been demonstrated that neural networks have the capacity needed to represent complex PDE solutions; however, the non-convex loss landscape imposed by the PDE/ODE constraint results in difficulties when achieving convergence to a high accuracy [2]. In this project, we will investigate methods for improving the accuracy of PINN ODE solutions while preserving computational efficiency. We will focus on sequential learning of ODEs by discretizing the integration timespan into sub-intervals, with a PINN approximating the solution within each interval. A particularly important aspect will be finding the “minimal” choice of hyperparameters leading to the best computational efficiency for a given problem.

The project requires an excellent mathematical background, and a willingness to explore beyond the material studied in your course. Experience with state-of-art machine learning is an advantage.

Access to departmental computational resources (GPU, CPU servers) is desirable although not a prerequisite for the project.

[1] L. Lu, X. Meng, Z. Mao, and G. E. Karniadakis, 'DeepXDE: A Deep Learning Library for Solving Differential Equations', SIAM Rev., vol. 63, no. 1, pp. 208–228, Jan. 2021, doi: 10.1137/19M1274067.

[2] A. Krishnapriyan, A. Gholami, S. Zhe, R. Kirby, and M. W. Mahoney, 'Characterizing possible failure modes in physics-informed neural networks', in Advances in Neural Information Processing Systems, 2021, vol. 34, pp. 26548–26560. Accessed: Jul. 25, 2022. [Online]. Available:

<https://proceedings.neurips.cc/paper/2021/hash/df438e5206f31600e6ae4af72f2725f1-Abstract.html>

Bruce, Paul Dr

Project no:	IC-MSCP01
Project title:	Shock wave - boundary layer interaction
Supervisor:	Dr Paul Bruce
Category:	Computational;Theoretical;Analysis;Survey
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Shock wave - boundary layer interaction (SBLI) is an important and complex flow feature encountered in practically all high speed vehicles and flow systems. This project builds on the material covered in your Aerothermodynamics course (AERO97011) with the aim of pushing the boundary of our understanding of SBLI. The scope of the project will focus on one or more of the following broad topic areas: (1) impact of surface flexibility on SBLI; (2) prediction of heat transfer in a hypersonic SBLI; (3) computational methods for modelling SBLI; (4) SBLI flow control.

Project no:	IC-MSCP02
Project title:	Computational study of SBLI
Supervisor:	Dr Paul Bruce
Category:	Computational
Software:	PyFR (have agreed support to be given by Peter Vincent and his group)
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Developments in computational tools using high order methods - such as PyFR - offer great potential for exploring the flow physics of unsteady compressible flows. However, the ability of such methods to accurately predict the properties of shock / boundary layer interactions (SBLI) is unproven. This project will build on the foundations of previous studies in the department to apply PyFR to a number of classical SBLI-specific test cases including one or both of the following:

- (a) a natural laminar flow (NLF) supercritical aerofoil,
- (b) an isolator for a ramjet engine.

Project no:	IC-MSCP03/E
Project title:	Hypersonic foldable Aeroshell for Thermal protection using ORigami (HATHOR), Aerothermal analysis – Project 1
Supervisor:	Dr Paul Bruce (Co-Supervisor Dr Matthew Santer)
Category:	Experimental: Ready-made experiment; Computational; Theoretical; Design
Software:	Abaqus, Matlab, OpenFOAM
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

To land a payload on an extra-terrestrial body with an atmosphere is an immense challenge. To achieve sufficiently small ballistic coefficients, large braking areas are required which are greater than the diameters of available rocket fairings. This necessitates the use of deployable structures. Interaction of the atmosphere with the aeroshell at hypersonic velocities leads to high thermal loads which in turn requires the design of adequate thermal protection systems (TPS). There are numerous other interconnected aspects of the aeroshell design which much be addressed.

In the Department of Aeronautics, we are developing the Hypersonic foldable Aeroshell for Thermal protection using Origami (HATHOR) concept. This consists of rigid folding TPS panels supported by a series of umbrella-like deployable ribs which lock upon deployment into a faceted aeroshell geometry. We are carrying out an analytical programme which will be validated by an experimental campaign consisting of wind-tunnel tests at both sub- and supersonic velocities, dynamic deployment tests, TPS structural testing, and others as appropriate.

Two projects are available, including 1 experimental project, to work on the structural and aero-thermal aspects, and their mutual interaction, of the HATHOR lander.

Students allocated one of these projects will join the HATHOR research team. The project is intrinsically related to the project of same name supervised by Dr Santer.

Project no:	IC-MSCP04/E
Project title:	Hypersonic foldable Aeroshell for Thermal protection using ORigami (HATHOR), Aerothermal analysis – Project 2
Supervisor:	Dr Paul Bruce (Co-Supervisor Dr Matthew Santer)
Category:	Experimental: Ready-made experiment; Computational; Theoretical; Design
Software:	Abaqus, Matlab, OpenFOAM
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

See project description above.

Project no:	IC-MSCP05/E
Project title:	Low-temperature ablation experiments for re-entry vehicle applications
Supervisor:	Dr Paul Bruce (Co-Supervisor Prof Koon-Yang Lee)
Category:	Experimental
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This multi-disciplinary research project will combine aerodynamics and materials science to study the behaviour of re-entry vehicles as they enter the Earth's atmosphere from space. These vehicles are responsible for the safe return of payloads (including astronauts!) from space and thus our ability to reliably predict and characterize their performance is essential. Ordinarily, the study of these vehicles necessitates the use of advanced (read: extremely expensive!) materials and wind tunnels to recreate the extreme conditions (very high temperatures) experienced during a real re-entry mission. The primary aim of this project is to establish a capability within Imperial to study the physics of re-entry vehicles without the need to use exotic materials or generate such extreme conditions. This will be achieved through experiments in Imperial's new Mach 5 Supersonic Wind Tunnel using a scale model of a re-entry vehicle made of the organic compound naphthalene. It is postulated that the unique properties of naphthalene – a safe and widely available material – will allow us to recreate some of the key physics of a real re-entry vehicle mission at a fraction of the cost and complexity of previous studies.

Project no:	IC-MSCP06/E
Project title:	Heat transfer analysis of a deployable heat shield
Supervisor:	Dr Paul Bruce
Category:	Experimental, Computational
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Currently, the maximum mass that can be landed on Mars (or other celestial bodies) is limited by the entry vehicle heat shield size, which is constrained by the space available in the launch vehicle fairing (4.5 m diameter for Ariane 5). Future inter-planetary exploration missions necessitate the development of entry vehicles which can fit inside the available launch vehicle and then deploy to a larger size once in space by mechanical deployment. This project complements ongoing work and expertise at Imperial to consider the design and optimisation of the heat shield for such a vehicle. The scope is flexible (negotiable), but will likely include: (1) Literature review to develop an understanding of the re-entry environment; (2) computational work in MATLAB to develop an engineering model of heat transfer to a complex-geometry heat shield; (3) infra-red-thermography experiments in the Imperial College supersonic wind tunnel for validation.

Project no:	IC-MSCP07
Project title:	Heat flux sensor for robust aerobraking missions
Supervisor:	Dr Paul Bruce (Co Supervisor Dr Sara Rodriguez Marinas, Airbus D&S Engineer)
Category:	Computational;Design;Analysis
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Aerobraking involves using a planet's atmosphere to slow down an incoming spacecraft to circularise its orbit minimising the need for retro-propulsion. Such manoeuvres offer the potential to reduce inter-planetary spacecraft mass and ultimately cost. However, the uncertainties associated with such operations can be significant, especially those that impact the heat flux experienced by the vehicle. On-board, real-time measurement of heat flux during aerobraking is a highly attractive option for reducing uncertainty and enabling a spacecraft to more efficiently perform a corridor control at periapsis. This project will involve a comprehensive review of relevant literature and modelling work using MATLAB in order to provide answers to the following questions: (1) what are the most promising candidate technologies for capturing (direct and indirect) heat transfer measurements in free-molecular flow and transitional flow regimes? (2) what are the characteristics (accuracy/reliability/response time) of such a heat transfer sensor? (3) what is the (quantifiable) advantage of implementing such a sensor on an aerobraking mission?

This project is in collaboration with Airbus Defence and Space and will be co-supervised by Sara Rodriguez Marinas (AOCS/GNC engineer in Airbus D&S).

Project no:	IC-MSCPB08/E
Project title:	Aero-structural analysis of a grid-fin
Supervisor:	Dr Paul Bruce (Co Supervisor David Hayes, MBDA)
Category:	Experimental
Software:	As a contingency (expts unavailable) this project could run as a computational version using one of more of: Star-CCM+, OpenFOAM, Abaqus.
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Grid-fins offer a robust and efficient method for the control and stabilisation of high-speed objects such as guided missiles and reusable rocket stages. The unique geometry of a grid-fin – which comprises multiple slender aerodynamic surfaces in close proximity – makes it susceptible to high-speed fluid-structure interaction, but this aspect of their performance has not been widely studied. This project aims to target this deficit in understanding by undertaking the following tasks: (1) design and build a scale model of a grid-fin for testing in Imperial's supersonic wind tunnel; (2) perform aerodynamic testing of the grid-fin model to validate (steady) aerodynamic performance predictions from CFD and literature; (3) perform (steady and unsteady) aero-structural measurements of the grid-fin model at supersonic conditions, focusing on the impact of flexibility on steady and unsteady aerodynamic performance; (4) validate results through comparison with predictions of aero-structural performance from an in-house coupled aero-structural (FEA/CFD) tool. This project builds on previous research into grid-fins in the Aeronautics Department at Imperial and will be co-supervised by David Hayes, an engineer in the Mechanical Future Concepts department at MBDA Systems.

Project no:	IC-MSCPB09
Project title:	Aerodynamics in high-performance sports
Supervisor:	Dr Paul Bruce (Co Supervisor John Hale)
Category:	Computational;Analysis;Literature
Software:	Star-CCM+
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project focuses on the important role of aerodynamics in the competitive high-performance sports of cycling and rowing. The project specification is broad and flexible (negotiable) depending on the interests of the selected student, who will ideally have a strong interest in cycling and/or rowing. This project aims to advance the state-of-the-art in sports aerodynamics and its relationship with athlete physiology and race strategy. The project will comprise some or all of the following tasks: (1) literature review to establish the state-of-the-art in technologies to minimise aerodynamic drag, together

with tools for the measurement and estimation of drag using data from real-world tests and wind tunnel experiments, reduced-order models, and CFD; (2) select a suitable methodology/tool/framework for predicting aerodynamic drag; (3) identify a small number of promising novel technologies and quantitatively assess/predict any anticipated performance benefit; (4) identify optimal strategies for improving practical (rapid-turnaround) techniques for drag analysis. The project will be co-supervised by John Hale – an elite athlete and engineer and founder of Hale Dynamics.

Buxton, Oliver Dr

Project no:	IC-MSCOB01
Project title:	Examination of SGS turbulence models from experimentally acquired multi-scale data
Supervisor:	Dr Oliver Buxton
Category:	Theoretical;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Large eddy simulations (LES) do not attempt to resolve the finest scales present in a turbulent flow (dissipative length scales) but model them from the scales that are resolved. These fine scales are thus referred to as the sub-grid scales (SGS). Previously acquired experimental data has simultaneous realisations of a large scale field of view, with a lower spatial resolution and three smaller scale fields of view with a higher spatial resolution that are located within the large scale field of view. This is therefore analogous to having an LES data set and simultaneous SGS datasets. This project therefore seeks to assess various turbulence models to their prediction of the SGS stresses, as well as taking the SGS datasets themselves and analysing the distribution of the turbulence model coefficients if they are computed directly from the SGS stresses themselves.

Project no:	IC-MSCOB02
Project title:	Development of a particle tracking algorithm
Supervisor:	Dr Oliver Buxton
Category:	Computational;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Small particles may be advected by turbulent flows, for example sand particles in a sandstorm or pollutant particles in an urban flow. The physics of the transport of larger particles through a turbulent flow are currently unknown and therefore further experiments are required. The classical technique to determine a particle's trajectory through a turbulent

flow is particle tracking velocimetry. Simply, this relies upon imaging either a single particle or a number of particles with three or more high-speed cameras from different perspectives. With use of a suitable spatial calibration it is then possible to reconstruct a three-dimensional volume from the images captured from the three cameras simultaneously, and hence to produce a time-resolved time history of the particle(s)'s position in space. This project will develop a particle tracking algorithm from scratch. If time permits a simple experiment will be performed, in which a single particle falls through a body of water, e.g. a measuring cylinder, on which the algorithm can be tested.

Project no:	IC-MSCOB03/E
Project title:	Wing - propellor interaction
Supervisor:	Dr Oliver Buxton (co-supervisor Dr Paul Bruce)
Category:	Experimental;#Theoretical
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

The transition to zero-emission, hydrogen-powered aviation opens up the possibility for new aircraft configurations/designs. One promising design is distributed propulsion, in which the aircraft wing has multiple propellers spread along the wing span. The enhanced air velocity over the wing from the propeller wash can be a means to increase the lift that is produced at a given cruising speed. However, the wash from the propeller is unsteady and far from a "clean" uniform flow. A particular challenge for distributed propulsion aircraft is the high-lift configuration since the propeller wash interferes with the slats/flaps. In this project we will construct a model of a wing/propeller and test it in the wind tunnel. We will look to measure the aerodynamic performance (mean and fluctuating lift/drag) of a wing in the high-lift configuration/propeller assembly. This will be supplemented with simple, potential flow modelling. Note that some preliminary design work will need to be taken out in the December - May period to ensure that the model is ready for testing during the main project period May - September.

Project no:	IC-MSCOB04/E
Project title:	Testing of aerofoils designed for Martian rotorcraft
Supervisor:	Dr Oliver Buxton (co-supervisor Dr Kostas Steiros)
Category:	Experimental;#Computational
Software:	StarCCM PyFR
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

In April 2021 Ingenuity became the first man-made object to complete an extra-terrestrial powered flight; on Mars. Mars has an extremely "thin" atmosphere in comparison to Earth, with a much reduced surface pressure (as well as a different chemical composition; consisting primarily of carbon dioxide). As a result flight on Mars occurs at "peculiar" (with

respect to Earth) conditions: a combination of high Mach number to extremely low Reynolds number. Current aerofoils have been designed (with over a century's experience) to operate efficiently in a typical terrestrial parameter space (in terms of Mach and Reynolds number combination). As a result our existing aerofoil designs are inefficient for Martian flight. At Imperial College London we are developing simulation-based approaches to optimising Martian aerofoils but this will require experimental data for validation. We will be commissioning our brand new supersonic wind tunnel later this year which has been designed to vary both Mach and Reynolds number independently, enabling us to approach typical Martian conditions. This project will develop our testing capabilities in this new supersonic wind tunnel with an eye towards producing high-quality data to help with our efforts to optimise Martian aerofoils. In particular we will aim to produce force and pressure measurements over specifically designed Martian aerofoil sections. These experimental results will be supplemented with results from CFD, including both RANS and DNS using StarCCM/PyFR (our in-house compressible DNS solver).

Cantwell, Chris Dr

Project no:	IC-ACMCC01
Project title:	High-performance software for modelling plasma kinetics in tokamaks
Supervisor:	Cantwell, Chris Dr
Category:	Computational
Software:	Nektar++ (www.nektar.info), In-house PIC code.
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Nuclear fusion technology has the potential to solve many of the worlds current energy needs in a safe and environmentally friendly way. However, designing a suitable reactor (a tokamak) which can harness energy is a major scientific and engineering challenge. Computational modelling is the key tool which will enable us to realise a working tokamak. One of the challenges is how to model the dynamics of plasma inside the reactor. Particle-in-cell (PIC) methods are often used to solve plasma kinetic models, particularly when tracking fast particle orbits, owing to their reduced computational cost compared to a fully Eulerian representation of the particle distribution function. Projects are available, aligned with on-going work with the UK Atomic Energy Authority, in extending an in-house PIC software tool. This includes integrating the code with Nektar++ to solve field variables, exploring the performance of semi-implicit methods, and optimising performance of the code on GPUs. These projects have a strong software engineering focus and knowledge of C++ is essential.

Project no:	IC-ACMCC02
Project title:	Catching performance regressions in scientific codes
Supervisor:	Cantwell, Chris Dr
Category:	Computational
Software:	Nektar++ (www.nektar.info), Python.
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Creating robust and efficient software for use in scientific research requires developers to adhere to a range of best practices to ensure correctness under the use-cases for which they are designed. Version control, unit testing and regression testing are commonly used to support this aim. For codes used in high-performance computing environments it is also critical to ensure new code developments do not adversely affect performance. The aim of this project is to create a platform for the automated performance-testing of Nektar++, a now widely-used high-performance spectral/hp element code, and integrate it with an existing continuous integration system. The system should provide feedback in the form of various quantitative performance metrics across multiple tests, as well as an overall “pass” or “fail”. Test history should be stored and plotted to show performance history over time. Experience with a Linux command-line environment is essential. Familiarity with Python, as well as an interest in software engineering, is also desirable.

Project no:	IC-MSCCC03
Project title:	Computational modelling of cardiac electrophysiology
Supervisor:	Cantwell, Chris Dr
Category:	Computational;#Numerical
Software:	Nektar++ (www.nektar.info), Python, PyTorch
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

Cardiac electrophysiology describes the propagation of electrical signals in the heart which leads to its coordinated contraction and pumping of blood around the body. A cardiac arrhythmia is a disorganisation of these wavefronts, often due to the presence of diseased tissue and are treated clinically by isolating or destroying the problematic regions of tissue. However, for complex arrhythmias, identifying the appropriate treatment targets is challenging. Electrically activity can be mathematically modelled by a reaction-diffusion PDE. Computational modelling has the potential to revolutionise treatment of these conditions, while application of data science techniques and machine learning to measurements of the electrical signals can provide new insight into calibrating these models for personalised diagnostics and treatment. Projects are available in the development, validation and application of these technologies which, while using techniques and software similar to that used for fluid dynamics, provides an interesting alternative to traditional aeronautics applications in the context of biological phenomena. Knowledge of C++ or Python is desirable.

Project no:	IC-MSCCC04
Project title:	Development of a cloud-based web portal for interrogating high-order finite element datasets
Supervisor:	Cantwell, Chris Dr
Category:	Computational
Software:	Nektar++ (www.nektar.info)
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

High-performance computing is now used extensively for simulating a broad range of physical processes from fluid dynamics and material properties through to chemical reactions and the behaviour of complex biological systems. The scientific challenge and financial cost of generating rich datasets on large supercomputers is substantial. While these are of great value to other scientists around the world, sharing terabytes of data is not practical. Researchers typically need to apply operations such as extracting subsets of the data, interpolating onto different grids, computing auxiliary quantities and computing isocontours of particular quantities of interest which only requires a fraction of the total dataset. This project will investigate and prototype a web-based system for the remote analysis of high-order finite element datasets using cloud computing resources. It will allow the user to stipulate analysis protocols, and visualise the result through a web interface, or download the analysis in an open data format for local visualisation. This project has a strong focus on software engineering and knowledge of Python and C++ is essential.

Project no:	IC-MSCCC05
Project title:	Predicting the severity of coronary stenoses with computational fluid dynamics (CFD)
Supervisor:	Cantwell, Chris Dr
Category:	Computational, Numerical
Software:	Nektar++ (www.nektar.info), Python
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

A coronary stenosis is the narrowing of a coronary artery due to the build up of plaques which reduces the flow of blood to the heart. Severe cases are life-threatening, since insufficient oxygen reaches the heart muscle and leads to the death of regions of heart tissue. The severity of stenoses is established by measuring the fractional flow reserve (FFR) which relates the pressure after the stenosis to the pressure before the stenosis. A value of $FFR < 0.8$ is severe and necessitates emergency surgery to insert a stent. This project will apply high-order 3D and 1D reduced order computational models to a novel angiogram reconstruction approach, which can be more easily obtained in the emergency room or where CT imaging is unavailable. The aim is to develop and validate a procedure to rapidly and cost-effectively evaluate the severity of a patient's stenosis.

Chernyshenko, Sergei Prof

Project no:	IC-MSCSC01
Project title:	Auxiliary functions and sum of squares of polynomials (SOS1)
Supervisor:	Chernyshenko, Sergei Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;
	MSc in Composites

This project is the first of a group of related projects that grew up from research aimed at solving the problem of turbulence using the technique involving auxiliary functions or functionals, and often utilizing another technique, usually referred to as Sum of Squares of polynomials (SOS). The other projects of this group have SOS2, SOS3 and so on in their titles.

Before considering further the projects of this group, have a look at the Introductory text at <https://www.imperial.ac.uk/aeronautics/fluidynamics/Projects/IntroToAuxiliaryFunctions.pdf> . Choose SOS projects only if you love mathematics.

There are several short videos about this group of projects, see <https://www.imperial.ac.uk/aeronautics/fluidynamics/ChernyshenkoResearch/misc.php> .

The students doing these projects will have a considerable freedom in choosing a particular subtopic, and this choice will be offered during the projects after the students learn more about this area. If you are interested, select any of these projects. All the students taking these projects will start with the same literature review and exercises, and then will branch into the particular area they will like most. This project is a part of the ongoing research in the department.

The SOS projects are suitable only to students with strong mathematical background, and they usually require using Matlab.

Sum of squares of polynomials technique is a recent breakthrough in mathematics, which allowed checking and constructing positive-definite multivariate polynomials in a computationally efficient way. The method proved to be highly efficient in many areas, in particular in control theory. In fluid mechanics, thanks to the introduction of auxiliary function approach, it offered new opportunities in global stability theory, flow control, and bounds on various quantities related to turbulent flows, for example lift and drag. This is a new and exciting area of theoretical research in fluid dynamics, and it finds applications well beyond.

SOS1 is a generic project covering all applications of auxiliary functions and SOS. The other SOS projects are examples of specific possible topics. All students doing SOS projects will have a choice of any related topic, including those not listed. Hence, choose any if you like one.

Project no:	IC-MSCSC02
Project title:	Finding orbits in space (SOS2)
Supervisor:	Chernyshenko, Sergei Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;
	MSc in Composites

This project is one of a group of related projects. Please read the SOS1 description first, and look at the links there. Note the strict requirement of being in love with mathematics!

This project aims to apply the auxiliary function method and the sum of squares of polynomials technique to the problem of finding nonstandard periodic orbits, which can be used by spacecraft exploring the Solar system (google for past missions utilizing the idea :-)). See “Finding extremal periodic orbits with polynomial optimisation, with application to a nine-mode model of shear flow” by Mayur Lakshmi et al, SIAM Journal on Applied Dynamical Systems 19(2):763-787, 2020, see here: <https://doi.org/10.1137/19M1267647>. The same technique will be applied to the system of equations describing the spacecraft motion.

One of the videos at <https://www.imperial.ac.uk/aeronautics/fluidynamics/ChernyshenkoResearch/misc.php> is about this project.

Project no:	IC-MSCSC03
Project title:	Background flow for Vlasov equations (SOS3)
Supervisor:	Chernyshenko, Sergei Prof
Category:	Theoretical
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;
	MSc in Composites

This project is one of a group of related projects. Please read the SOS1 description first and look at the links there. Note the strict requirement of being in love with mathematics!

Vlasov equations are a system of equations governing the dynamics of rarefied plasma in nuclear fusion devices like tokamaks, in electric propulsion thrusters used in satellites, and in other applications. Background flow method is an auxiliary functional method in which the auxiliary functional is a sum of the energy and an inner product of its argument with a vector field called a background flow. The background flow method is the current workhorse of the methods of bounding time averages, but it was never applied to Vlasov equation. Current research in the department aims to solve a chain of similar problems, the simplest of which has already been solved, and the latest will be the full Vlasov equation case. The student doing this project will join at the stage to be achieved by the start of the project. For the first steps see the first part of Ryo Haga FYP report:

https://www.imperial.ac.uk/aeronautics/fluidynamics/Projects/Ryo_HAGA_FYP_section.pdf

Before applying for this project consider if you would like and able to do such work.

This project is purely analytical, pen-and-paper mathematical study. Also note the practical importance of nuclear fusion and space propulsion.

Project no:	IC-MSCSC04
Project title:	Bounds for mean parameters of a shear layer described by a reduced-order model (SOS4)
Supervisor:	Chernyshenko, Sergei Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;
	MSc in Composites

This project is one of a group of related projects. Please read the SOS1 description first and look at the links there. Note the strict requirement of being in love with mathematics!

The goal of this project is to find out the differences in applying the auxiliary function method to wall-bounded flows and free shear layer flows. This will be done by finding bounds for the reduced order model of the shear flow proposed by A.Cavalieri et al. (<https://doi.org/10.1017/jfm.2021.1029>) and comparing the observations with observations for wall-bounded flow in section 5.2 in M. Lakshmi et al. (<https://doi.org/10.1137/19M1267647>).

This project will require the use of the MATLAB toolbox YALMIP (see <https://yalmip.github.io/tutorial/basics/> for what this involves).

Project no:	IC-MSCSC05
Project title:	Auxiliary functions for systems with noise (SOS5)
Supervisor:	Chernyshenko, Sergei Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;
	MSc in Composites

This project is one of a group of related projects. Please read the SOS1 description first and look at the links there. Note the strict requirement of being in love with mathematics!

The bound for a time average obtained using the auxiliary function method applies for all solutions. For example, usually the drag in a laminar flow is much smaller than the drag in a turbulent flow. This means that the lower bound for the drag cannot be greater than the drag in the laminar flow. Such a bound can be too far away from the actual drag in the turbulent flow. In practice, at high Reynolds number the laminar flow is unstable and the ever-present small noise will make it irrelevant. The auxiliary function method can be extended to systems with noise. This project will aim to investigate the behaviour of the auxiliary function as the noise intensity tends to zero. This can be done in three ways. The best possible auxiliary function satisfies a second-order linear partial differential equation, with the noise intensity being a coefficient at the highest derivatives. There are three ways to explore. First, the auxiliary function can be found using SOS optimization, and its dependence on noise intensity analyzed. (This will require the use of the MATLAB toolbox YALMIP. See <https://yalmip.github.io/tutorial/basics/> for what this involves). However, the range of noise intensities will be limited by available computational resources. Second, a specialist code can be written for solving the equation for the auxiliary function. This will require knowledge of a high-performance language like Fortran or C and knowledge of numerical methods for PDE. Third, a pen-and-paper analysis can be done using asymptotic methods. To do this, the student will need either to know these methods or to learn them.

The progress made so far in this topic is exemplified by the paper by Fantuzzi et al. (2016) available at <https://epubs.siam.org/doi/pdf/10.1137/15M1053347>

Project no:	IC-MSCSC06
Project title:	Bounds for mean parameters of chaotic regimes in the models of ionic currents in biological membranes (SOS6).
Supervisor:	Chernyshenko, Sergei Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;
	MSc in Composites

This project is one of a group of related projects. Please read the SOS1 description first and look at the links there. Note the strict requirement of being in love with mathematics!

Models of ionic currents are routinely used, for example, in cardiac arrhythmia research. The project will start with applying the auxiliary function techniques to the famous Hodgkin-Huxley model, for the development of which Alan Hodgkin and Andrew Huxley received the

1963 Nobel Prize in Physiology or Medicine. It will then progress to more complicated models. The goal of the project is exploratory: to see how the technique applies to models of this type.

This project will require the use of the MATLAB toolbox YALMIP (see <https://yalmip.github.io/tutorial/basics/> for what this involves).

Project no:	IC-MSCSC07
Project title:	Bounds for mean reaction rates of chaotic chemical reactions (SOS7)
Supervisor:	Chernyshenko, Sergei Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;
	MSc in Composites

This project is one of a group of related projects. Please read the SOS1 description first and look at the links there. Note the strict requirement of being in love with mathematics!

The Belousov–Zhabotinsky reaction in a continuous-flow stirred-tank reactor is the best-studied chaotic chemical system. The project will start from finding bounds for reaction rate in a simple mathematical model of this system, and then proceed to more complex reactions. The goal of the project is exploratory: to see how the technique applies to models of this type.

This project will require the use of the MATLAB toolbox YALMIP (see <https://yalmip.github.io/tutorial/basics/> for what this involves).

Project no:	IC-MSCSC08
Project title:	Accelerating time averaging (SOS8)
Supervisor:	Chernyshenko, Sergei Prof
Category:	Computational;#Theoretical;#Numerical
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;
	MSc in Composites

This project is one of a group of related projects. Please read the SOS1 description first and look at the links there. Note the strict requirement of being in love with mathematics!

In many practical applications the main goal of numerical modeling of a dynamical system is to obtain the long-time average of a certain set of parameters, such as, for example, the lift and drag force acting on an aircraft. The solutions of large nonlinear systems of equations often exhibit a chaotic behaviour. When the amplitude of the fluctuation of the quantity of interest is large, or when a very high accuracy is required because the effect to be investigated is small, obtaining time-averaged quantities with sufficient accuracy requires large averaging intervals, and thus, expensive numerical calculations.

The recently proposed method of accelerating the averaging amounts to replacing the quantity being averaged with another quantity having the same average but fluctuating less. This is achieved using some of the ideas of the auxiliary function method. Namely, one of

the key elementsof that method is that for any differentiable function $V(x)$, where x is the vector describing the state of the dynamical system, the infinite time average of $dV(x(t))/dt$ is zero provided that the trajectory of the dynamical system is bounded, which is always the case when infinite time averaging is meaningful. Hence, rather than numerically averaging the quantity of interest, which we will denote F , one can average $F+dV/dt$, for any V . The function $V(x)$ can be represented as a linear combination of a finite set of a suitably chosen basis functions. The coefficients of the linear combination can then be optimized to minimize the fluctuation of the finite-time average approximation of the infinite-time average.

The project is suitable for several students, each applying the method to a different system. The work will amount to numerically modeling the system of choice and then accelerating the averaging by the method described above. The necessary skills expected from the student are mathematics and programming. The project can be done in Matlab, but knowing a high-performance language would be an advantage for big systems.

One of the videos at

<https://www.imperial.ac.uk/aeronautics/fluidynamics/ChernyshenkoResearch/misc.php> is about this project.

Doorly, Denis Prof

Project no:	IC-ACMDJD01
Project title:	Random walk methods for the simulation of diffusion and perfusion – Project 1
Supervisor:	Doorly, Denis Prof
Category:	Theoretical/Computational/numerical
Software:	Matlab and Python
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

(Collaboration with Dr. A Scott, CMR physics).

The project will investigate methods to improve the modelling of diffusion across semi-permeable membranes, using random walk methods. It will extend previous projects in this area and the work of Rose et al: Novel insights into in-vivo diffusion tensor cardiovascular magnetic resonance using computational modelling and a histology-based virtual microstructure.

Pre-requisite: Good understanding of undergraduate mathematics. Some background to the application area: <https://doi.org/10.1002/mrm.27561>; **some theoretical background:** <https://royalsocietypublishing.org/doi/10.1098/rspa.1992.0025>

Project no:	IC-ACMDJD02
Project title:	Random walk methods for the simulation of diffusion and perfusion – Project 2
Supervisor:	Doorly, Denis Prof
Category:	Theoretical/Computational/numerical
Software:	Matlab and Python
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

See project description above.

Project no:	IC-ACMDJD03
Project title:	Simulation of a quantum computer – Project 1
Supervisor:	Doorly, Denis Prof
Category:	Experimental;Numerical
Software:	Matlab, Anaconda (Python)
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Simulation of a quantum computer: The project will investigate methods to simulate the execution of an algorithm on a quantum computer. A starting point could be an existing matlab package - see <https://fr.mathworks.com/matlabcentral/fileexchange/73035-quantum-computer-simulator>

Project no:	IC-ACMDJD04
Project title:	Simulation of a quantum computer – Project 2
Supervisor:	Doorly, Denis Prof
Category:	Experimental;Numerical
Software:	Matlab, Anaconda (Python)
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

See project description above.

Project no:	IC-ACMDJD05
Project title:	Aerosol transport in the airways
Supervisor:	Doorly, Denis Prof
Category:	Experimental; Numerical; Computational
Software:	Matlab, StarCCM
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

The project will apply computational modelling to study aerosol transport in the upper airways. If experimental work resumes as normal, it may be possible to perform some experiments in an experimental project.

Hewson, Robert Dr

Project no:	IC-MSCRH01
Project title:	Firedrake Fluid Dynamics
Supervisor:	Hewson, Robert Dr
Category:	Computational;#Numerical
Software:	Python and Firedrake - supported locally.
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Firedrake is an open source python package which allows for the efficient and parameterisable solution of a range of problems. This project will focus on the development of a flexible set of tools written in python and Firedrake to solve a range of Navier-Stokes fluid flow problems. The addition of turbulence, compressibility and other effects, along with the potential to use advanced GPU resources makes this a challenging project for someone who enjoys programming.

By the end of this project, you will have demonstrated how Firedrake can be used to simulate a range of external flows in a flexible and benchmarked way, and demonstrated the speedups achievable with GPU and HPC resources.

Project no:	IC-MSCRH02
Project title:	Stressed Topology Optimisation
Supervisor:	Hewson, Robert Dr
Category:	Computational;#Numerical
Software:	Python and Firedrake - supported locally.
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

The use of topology optimisation to derive structures is a well established method. However, it tends to be dominated by maximising the structural stiffness. This is often useful, but stress is often equally important.

The challenge with including stress is that it is often only the peak stress which is of importance, and needs minimising or constraining. This proves difficult in topology optimisation, where integral constraints and objectives are often applied. This often leads to "constraint aggregation" which leads to less precise and smeared constraints being introduced. There are numerous other challenges which need to be considered which make stress control challenging in TO.

We will address some of these challenges by introducing a "slack design variable" for stress, allowing the precise local calculated stress values to be determined.

This project will involve a lot of programming in python and the use of the Firedrake FE package. Python experience is not essential, but you should enjoy programming and have experience of debugging code.

Project no:	IC-MSCRH03
Project title:	Multidimensional, multiscale structural shell design
Supervisor:	Hewson, Robert Dr
Category:	Computational;#Numerical; Design
Software:	Python and Firedrake - supported locally.
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project will focus on the design, through the application of topology optimisation, the design of small scale periodic structures for application in shell design. See <https://www.imperial.ac.uk/structural-metamaterials/research/shell-metamaterials/> for examples of a multiscale shell structure where this project will focus on the design of the small scale structure.

The small scale (semi-) periodic structural component will be optimised for a range of different requirements, typically maximising the terms of the classic A, B and D matrices used to describe the shell behaviour (used widely in classic lamination theory) as well as out

of plane displacement, to facilitate morphing of the shell as well as activation of surface roughness.

Obvious applications include the aerodynamic morphing of a structure with changes in roughness included in the gross shape morphing design.

This project will be computational in nature and use python and the firedrake FE package. You do not need experience in using python, but it would be advantageous, and you absolutely must enjoy programming and debugging code!

Project no:	IC-MSCRH04
Project title:	Flexible FSI in Firedrake
Supervisor:	Hewson, Robert Dr
Category:	Computational;#Numerical; Design
Software:	Python and Firedrake - supported locally.
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project will focus on fluid-structure interaction modelling using the python based FE package Firedrake.

The NS equations will be used to describe the fluid problem and a high fidelity linear elastic problem will be used to describe the structural problem. There is some flexibility regarding the problem we will look at, but the aero-elastic response of a wing is one such problem we could consider.

You do not need experience of using python to undertake this proeject, though it would be advantageous. However, you should enjoy programming and debugging your code.

Project no:	IC-MSCRH05
Project title:	Multiphase fluid flow in Firedrake
Supervisor:	Hewson, Robert Dr
Category:	Computational;#Numerical; Theoretical
Software:	Python and Firedrake - supported locally by research group.
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project is concerned with using the open source finite element package FireDrake to model multiphase fluid flow problems, including the role of moving contact lines and the singularities which arise. This has applications in modelling how fluid can be transferred during the coating and printing processes present in the manufacture of photovoltaic cells and a range of other applications.

Firedrake is a package in python so you should be happy programming and debugging code (though not nescessarily in python).

Project no:	IC-MSCRH06
Project title:	How do lubricated surfaces touch?
Supervisor:	Hewson, Robert Dr
Category:	Computational;#Numerical; Theoretical
Software:	Python
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

The challenge of modelling lubricated contacts as the gap between opposing surfaces tends to zero is one which has proved to be a significant challenge for many years. Without surfaces coming into contact there is no wear of surfaces, yet we know wear occurs in real tribological contacts. As gaps tend to zero the pressure in the contact tends to infinity, forcing the contact apart. We will study ways to overcoming the apparent contradiction arising from the no-slip condition (and even in most slip models) as surfaces move together. We will take inspiration from models of dynamic contact angles and deliver a model for the transition from a fluid film separated geometry to that of contact.

This project will most likely use python and will also have a theoretical component.

Project no:	IC-MSCRH07
Project title:	Aerodynamic shape optimisation
Supervisor:	Hewson, Robert Dr
Category:	Design;#Numerical
Software:	Python and Firedrake. Supported locally by me and my group.
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project will develop methods for aerodynamic shape optimisation. At the core of this work will be the finite element solver, Firedrake, which will be used to solve the Navier-Stokes equations. This is an open source finite element package which we will use to efficiently solve the NS equations before using its automatic differentiation capability to solve the design sensitivities - a key component of an efficient gradient optimisation method. There is lots to explore, from the inclusion of turbulence models, compressibility, three dimensionality. In the first instance we will use my research group's workstation, a 24 core, >600GB RAM number cruncher, and if we require more power we will turn to the college HPC facility.

If you want to do this project you should enjoy programming, python is not essential, but an ability to debug and tweak code is! You will gain experience of optimisation methods, computational mechanics and potentially using GPUs to accelerate computational mechanics. In addition to your supervisor, you will interact with PhD students and postdocs in my group.

Project no:	IC-MSCRH08
Project title:	Level set topology optimisation
Supervisor:	Hewson, Robert Dr
Category:	Computational;#Design;#Numerical
Software:	Python and firedrake - supported locally.
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Conventional topology optimisation uses a pseudo-density to arrive at an optimum topology. This is what is typically used in commercial packages and is often termed, mainly for the benefit of product designers, generative design. However, other techniques exist to describe a topology, one of these being the level set approach where spatially varying scalar field's zero iso-level is used to describe the boundary between material and void. There are a number of approaches to this method, but the one we will explore is the use of multiple optimised levels summed together in an inverse homogenisation approach (trying to maximise certain terms in a homogenised stiffness matrix) to allow optimal periodic structures to be produced which can evolve spatially without losing connectivity between designs.

This is a conceptually challenging thing to get to grips with, but will require some programming in python and use of the python package firedrake. You should be happy programming and debugging your own code.

Hwang, Yongyun Dr

Project no:	IC-ACMYH01
Project title:	Data-driven modelling and control of turbulent flow
Supervisor:	Hwang, Yongyun Dr
Category:	Computational;#Theoretical
Software:	Matlab
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods;

The project is aimed to examine novel technique that construct a nonlinear dynamical system using the data obtained by direct numerical simulation. The direct numerical simulation will first be performed to obtain the data associated with the dynamics of coherent structures (i.e. highly organised motions) in a wall-bounded turbulent shear flow. Using this data, a set of optimisation will be formulated to generate a model in the form of systems of ODE. Finally, the constructed model will be studied in detailed comparison with the original direct numerical simulation data. The project is both theoretically and computationally demanding, and it enables the student to learn the theories of chaos and turbulence.

Project no:	IC-ACMYH02
Project title:	Quasi-linear approximation of turbulent flow
Supervisor:	Hwang, Yongyun Dr
Category:	Computational;#Theoretical
Software:	Matlab
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods;

Quasi-linear approximation is a new approach that simulates only the essential part of dynamics of flows. The key idea is to take only important part of nonlinearity, while neglecting all the other nonlinear terms in analysing flows. The goal of this project is to test this idea using in-house numerical simulation tools. Strong mathematical capability and interest in scientific computation are required.

Project no:	IC-ACMYH03
Project title:	Instabilities and pattern formation in active fluids
Supervisor:	Hwang, Yongyun Dr
Category:	Computational;#Theoretical
Software:	Matlab
Confidential:	Yes
Offered to:	
	MSc in Advanced Computational Methods;

Active fluid refers to fluid in which active agents such as self-propelling particles and swimming microorganisms are suspended. In such a fluid, energy input to the system is given by the active agents, leading to fascinating instabilities and pattern forming motions that have not been observed in normal fluids. In this project, the student will study instabilities and pattern formation arising in such a system.

Project no:	IC-ACMYH04
Project title:	Stochastic fluid dynamics and control
Supervisor:	Hwang, Yongyun Dr
Category:	Computational;#Theoretical
Software:	Matlab
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods;

Noise is very common to environment: for example, in aircraft wings, it is common to see gust which can sometimes cause fatal accidents. In general, the effect of noise on fluid systems has often been studied in laminar/turbulent transition. Here, we will extend this notion to more general fluid system and study how the noise affect the bifurcation and the instability and coherent structure dynamics.

Kerrigan, Eric Prof

Project no:	IC-MSCEK01
Project title:	Dynamic optimization using the Julia programming language
Supervisor:	Prof Eric Kerrigan (with Eduardo Vila)
Category:	Computational;#Theoretical;#Analysis;#Numerical
Software:	Matlab Julia (open-source, freely downloadable)
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Nearly all optimal control and estimation problems, as well as many system design problems, can be formulated as dynamic optimization problems. These are also often called trajectory optimization problems in the aerospace or robotics communities, where one typically wants to get a vehicle or object from one point to another point in space, while minimising time, energy or cost.

In dynamic optimization one seeks to optimize an objective over functions of time and space, subject to differential equations (arising due to modelling the dynamics) and inequality constraints (arising due to safety, performance or physical considerations), possibly while providing guarantees despite modelling uncertainties, disturbances and noise. These problems can be solved efficiently using tailored, structure-exploiting finite-dimensional numerical

methods. For an introduction to numerical methods for dynamic optimization, see doi.org/10.1137/16M1062569 and doi.org/10.1137/1.9780898718577

Julia is a modern high-level language, which employs sophisticated ideas from computer science to allow for rapid algorithm development (similar or easier than Matlab or Python) with an emphasis on high performance (comparable or faster than C). Julia might just one day overtake C/C++ as the language of choice for high-performance scientific computing. See julialang.org and the paper doi.org/10.1137/141000671 for more details. If you have not yet heard of Julia, you will definitely hear a lot about it in the future!

This project will involve contributing towards the development of a user-friendly and efficient package for dynamic optimization in the Julia Language. The aim is to replicate and extend the state-of-the-art methods currently implemented in the Matlab-based package ICLOCS2, developed by my group (Imperial College London Optimal Control Software - see www.ee.ic.ac.uk/ICLOCS/).

Ideally, the student should have some experience in optimization and control theory. The student has to be familiar with more than one programming language and have done object-oriented programming. **DO NOT TAKE THIS PROJECT IF YOU ONLY KNOW MATLAB.**

This project can be taken by more than one student. If there is more than one student assigned to this project, they will be assigned separate tasks from other students, but will be expected to collaborate with other members of the team.

We already have some MEng and MSc projects contributing to this project, hence the list of open tasks is constantly evolving. However, the following list gives an indication of possible tasks that any new student could choose to focus on:

1. Automatic differentiation in optimal control: Interfacing JuliaDiff to dynamic optimisation problems.
2. Collocation methods: Implementation of different collocation methods to solve optimal control problems using existing packages for polynomial approximation
3. Runge Kutta methods: Implementation of different Runge Kutta methods to solve optimal control problems.

Each student assigned to this project should also choose one or more engineering case studies to which they have to apply and test their code. This list is also constantly evolving, but a list of possible case studies include:

1. Optimal ventilation of patients (e.g. patients with COVID-19),
2. Formation flight of UAVs,
3. Rocket launch and landing ,
4. Spacecraft rendezvous and docking,
5. Aerial communication networks,
6. Energy management in a building,
7. Racing car laptime optimization.

Project no:	IC-MSCEK02
Project title:	On-line learning and optimization of controller performance for planetary rovers
Supervisor:	Prof Eric Kerrigan
Category:	Computational;#Theoretical;#Design;#Analysis;#Numerical
Software:	Matlab and Simulink/Simscape
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Outer space and planetary environments are extremely harsh – substantial temperature gradients of different frequencies, dusty environments, unintuitive impacts of low gravity and/or vacuum and effects of solar wind are just few examples. Degradation of mechanisms is common and unavoidable. Moreover, the application field requires solutions that are zero-maintenance and can work multiple (Earth) years.

Within this project we are looking for ways to counteract such degradations that impact driving mechanisms of planetary rovers. Planetary rovers (like ExoMars [https://www.esa.int/Science_Exploration/Human_and_Robotic_Exploration/Exploration/Exo Mars](https://www.esa.int/Science_Exploration/Human_and_Robotic_Exploration/Exploration/Exo_Mars) or SFR https://www.esa.int/ESA_Multimedia/Videos/2020/02/Sample_Fetch_Rover_for_Mars_Sample_Return_campaign) utilise multiple driving and steering actuators that are exposed to those harsh conditions and its performance will vary in time. Control of those actuators to execute body-level commanding is done by low-level controllers (typically a set of PID controllers to control lateral and longitudinal errors), which are sub-systems of the rover's trajectory control system. The objective of this project is to develop a solution to monitor the performance of those controllers and adjust their parameters to achieve best results, while respecting actuator limitations and minimising the impact of wear on them.

Essential: The student should have completed a course on state-space methods for control system design and be proficient with Matlab.

Desirable: Experience with Simulink/Simscape is desirable.

This project is in collaboration with Airbus Defence and Space, but will be carried out in the Department of Aeronautics. However, the student will be expected to interact with and deliver regular progress updates to Airbus engineers.

Project no:	IC-MSCEK03
Project title:	Distributed numerical optimization for trajectory planning of unmanned air vehicles
Supervisor:	Prof Eric Kerrigan (with Lucian Nita)
Category:	Computational;#Theoretical;#Design;#Analysis;#Numerical
Software:	Julia
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project will focus on derivative-free methods for real-time predictive control using the Julia programming language.

Predictive control is the most widely implemented advanced control technique in industry. Predictive control is a technique that relies on solving an optimization problem repeatedly in order to compute the desired control action. However, as the size of the problem grows, it becomes increasingly more difficult to solve it in real-time. Imagine a large network of agents that need to coordinate with each other. To perform a desired task, each agent is running estimation and control algorithms on the on-board computer. The questions that naturally arise are: Can existing algorithms be adapted to cooperatively find a single solution to the desired problem? In any application, constraints on computational time play a critical role - how should a new optimization approach should look like to be able to achieve the best performance in the given time? If the time limit is very short, agents can execute the best control action that can be obtained for a fixed amount of computation and improve it the next time the algorithms are executed. This forms the basis of machine learning and numerous models are constructed every day based on data. The aim of this research is to generate acceptable solutions quickly and improve them as new data comes in.

The main objective of this project is to explore the applicability of recent, state of the art derivative-free methods for designing distributed optimization algorithms and develop a new package for distributed optimization that considers the computational time constraints. This project will build on the past work of the group that used gradient-based methods for a trajectory-planning problem involving a network of UAVs, as well as the internally developed DirectSearch.jl package (<https://github.com/ImperialCollegeLondon/DirectSearch.jl>).

Essential: The student should have taken a course on state space methods for control systems.

Ideally, the student should have an interest in numerical optimization and be familiar with programming in at least two different languages. **DO NOT TAKE THIS PROJECT IF YOU ONLY KNOW MATLAB.**

Project no:	IC-MSCEK04
Project title:	Advanced gyro-less spacecraft attitude estimation
Supervisor:	Prof Eric Kerrigan
Category:	Computational;#Theoretical;#Design;#Analysis;#Numerical
Software:	Matlab and Simulink/Simscape
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Modern space missions require high levels of accuracy in the attitude estimate, so that fine pointing is achieved and payload data is post-processed with sufficient precision. For this reason several missions employ high-precision gyroscopes which complement star tracker measurements via data fusion. Accurate gyroscopes are however notoriously expensive, therefore efforts towards gyro-less attitude estimators have been made by industry and research (see <https://doi.org/10.2514/2.4071>). For high-precision applications, effects such as structural flexibilities (due to solar arrays or instrument booms) and model uncertainty play a significant role in the degradation of the attitude estimate. On the other hand, advanced techniques such as machine learning and dynamic programming have shown promising results in control and estimation tasks (e.g. <https://doi.org/10.1109/TNN.2005.863458>).

The aim of the project is to evaluate the effectiveness of modern estimation techniques in the frame of gyro-less attitude estimation. An algorithm for attitude estimation will be developed and tested on the defined use case, starting from the model of a spacecraft equipped with star tracker and reaction wheels, The use of all available signals is advised: star tracker attitude measurement, reaction wheels commanded torques, reaction wheel speeds and any other useful signal.

Essential: The student should have completed a course on state-space methods for control system design and be proficient with Matlab.

Desirable: Experience with Simulink/Simscape is desirable.

This project is in collaboration with Airbus Defence and Space, but will be carried out in the Department of Aeronautics. However, the student will be expected to interact with and deliver regular progress updates to Airbus engineers.

Project no:	IC-MSCEK05
Project title:	Constraint handling for scenario-based optimization
Supervisor:	Prof Eric Kerrigan (with Lucian Nita)
Category:	Computational;#Theoretical;#Design;#Analysis;#Numerical
Software:	Matlab and Simulink/Simscape
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project will focus on distributed constraint handling for large-scale scenario-based optimization and will be developed in the Julia programming language.

Robust predictive control has received a lot of attention recently, both from industry and academic researchers. This is one of the most popular and studied topics in the field of control engineering with many practical applications ranging from fail-safe systems, energy microgrids, temperature control inside buildings, to indoor UAV trajectory planning, self-driving cars, and many others. To ensure constraint satisfaction in real time under unknown disturbances we generate multiple scenarios based on different disturbance realizations and enforce them as constraints during the optimization process. However, as the number of sample scenarios grows, only a very small fraction of the added constraints will be active. The questions that need to be answered in this project are: Which constraints need to be included in the optimization problem? How best to find the most violated constraints? How best to use the computational resources available in order to obtain a feasible solution? How will computational time constraints will affect the quality of the solution? Since data-based models are just only approximate the physical behaviour of the system and the approximation error can be treated as an unknown disturbance, the above stated questions are also relevant within the field of safe learning in control. The aim of this research is to derive a policy for adding and removing constraints at each iteration and in order to solve a large-scale constrained optimization problem.

The main objective of this project is to explore the applicability of recent, state of the art constraint handling methods for designing large -scale scenario -based predictive controllers. This project will build on the past work of the group, which implemented an external active set method and a parallel sorting algorithm for ordering the constraint set based on the existent violation. (See <https://github.com/ImperialCollegeLondon/External-Active-Set/tree/main/src>) A numerical example will needs to be implemented in order toto demonstrate the applicability of the newly developed method. In the numerical simulation we aim to determine the optimal input sequence for a path planning problem of UAV swarms.

Essential: The student should have completed a course on state space methods for control system design.

Ideally, the student should have some background in optimization and be familiar with programming in at least two different programming languages. **DO NOT TAKE THIS PROJECT IF YOU ONLY KNOW MATLAB.**

Knoll, Aaron Dr

Project no:	IC-AAEAK01/E
Project title:	Water as a propellant for spacecraft electric propulsion
Supervisor:	Knoll, Aaron Dr
Category:	Experimental
Software:	Ni Labview Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

Electric propulsion is becoming the state-of-the-art propulsion method for telecommunication, exploration and science satellites. The higher efficiency of the engines allows the same payload to use a smaller spacecraft bus or for very large payloads to be fitted onto launchers that would otherwise be impossible. The most prevalent engine and

propellant combination used today are hall effect thrusters using Xenon. Xenon is a very expensive and rare propellant, which has led to a drive in recent years to move to an abundant and low-cost alternative.

This experimental study will investigate the use of water vapour as a propellant for Hall Effect Thrusters. A prototype thruster, called the “WET-HET”, will be used for these experiments. This thruster has been tailored to operate on lightweight molecular propellants and its performance has previously been characterized using oxygen propellant in a PhD research activity. The aim of this project is to characterize the thruster performance using water vapour, and compare the results to simulations conducted as part of a previous MEng project from 2020/21.

Project no:	IC-AAEAK02/E
Project title:	Construction of a 3-axis Langmuir probe instrument for plasma thruster plume characterization
Supervisor:	Knoll, Aaron Dr
Category:	Experimental
Software:	Autodesk Fusion 360 LabView
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

A Langmuir probe instrument is used to directly measure physical properties of a plasma including density, temperature and potential (voltage). To understand/verify the underlying physics of operation of plasma thrusters, it is useful to profile the plasma parameters of the thruster plume over a 3-dimensional volume. The purpose of this project is to construct a 3-axis translating Langmuir probe instrument capable of surveying the plasma properties with minimal disruption to the plasma thruster discharge. The translation mechanism will operate within a vacuum chamber environment, and interface with an external PC via LabView. This project builds on the design that was developed during an MSc project in 2019/20. It is anticipated that this instrument will play an essential role in future experimental activities conducted within Imperial's new Plasma Propulsion Lab.

Project no:	IC-AAEAK03/E
Project title:	Experimental testing of a Hollow Cathode for small satellite applications
Supervisor:	Knoll, Aaron Dr
Category:	Experimental
Software:	NI LabVIEW
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

Hollow cathode neutralizers are an essential component of almost every electric propulsion (plasma thruster) technology on the market. The purpose of the hollow cathode is to neutralize the outgoing ion beam to prevent charge accumulation on the satellite and, in the

case of Hall Effect Thrusters, to provide the electron current to sustain the discharge. Despite their relative importance, these devices continue to be the 'weak link' in terms of the reliability and cost of the overall system. Additionally, there are virtually no hollow cathodes on the market that meet the needs of the growing small satellite industry. This project builds on a design that was developed and constructed during a MSc project in 2020/21. The purpose of this current project is to demonstrate its successful operation within the large vacuum chamber facility (Boltzmann Chamber) and to characterize its performance. No prior plasma physics experience is necessary, but I'm looking for students with an openness to learn new things. Through this project you will learn many practical skills that are specific to plasma thrusters, so it's an ideal opportunity for those interested in becoming involved in the field of spacecraft electric propulsion.

Project no:	IC-ACMAK04
Project title:	Taming plasma turbulence using unsteady electrostatic forcing
Supervisor:	Knoll, Aaron Dr
Category:	Computational
Software:	PlasmaSim (in-house Particle-in-Cell plasma simulation in C++) Matlab
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Instability driven oscillations (turbulence) is inherent in virtually all plasmas, from the scale of the sun down to low power laboratory plasma sources. It is widely acknowledged that turbulence plays a central role in the macroscopic dynamics of a plasma, driving processes such as charged particle transport across magnetic fields. Understanding the origin and impact of plasma turbulence to natural and artificial plasmas is essential to predicting their behaviour. While there has been considerable progress made toward understanding instabilities in plasma, there has so far been limited success in tailoring these modes. This project has the ambitious aim to simulate the active control of turbulence within a magnetized plasma source using a novel method known as unsteady forcing. This method involves exciting the plasma with an external oscillating signal at a specific frequency to promote harmonic energy transfer between different naturally occurring modes/instabilities. The simulations will be conducted using PlasmaSim, a Particle-in-Cell (PIC) code that was developed in-house within the Aeronautics department (C++ language).

Project no:	IC-AAEAK05
Project title:	Design of a compact plasma toroid generator
Supervisor:	Knoll, Aaron Dr
Category:	Theoretical;#Design;#Analysis
Software:	Autodesk Fusion 360
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

The pursuit of fusion energy through magnetic plasma confinement is an enduring challenge for humankind. While generating the required temperatures and pressures needed to produce fusion energy is theoretically possible, natural instabilities and turbulence occurring within the plasma allow the high energy particles to escape the magnetic field and rapidly drain energy from the system.

The goal of this project is to design an experimental test-bed to study the turbulence within a magnetically confined plasma, and investigate possible strategies to control (tame) this turbulence. This project will involve the design of a generator for a compact plasma toroid – a self-stable magnetically confined plasma structure, with a donut-like shape. The goal will be to design a low-cost small-scale device that can be subsequently constructed and tested within the large vacuum test facility (Boltzmann Chamber).

Project no:	IC-ACMAK06
Project title:	Particle-in-cell simulation of a high power (20 kW) Hall Effect Thruster using water propellant
Supervisor:	Knoll, Aaron Dr
Category:	Computational
Software:	PlasmaSim (in-house Particle-in-Cell plasma simulation in C++) Matlab
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

This project considers a novel high-power Hall Effect Thruster (HET) that operates on water propellant. Hall Effect Thrusters are one of the leading technologies used for satellite propulsion today. In almost all cases, Xenon is used as the propellant gas. However, the demand for electric propulsion is rapidly exceeding the global production rate of Xenon leading to extremely high prices and market volatility of this rare material. We are currently developing a new type of Hall Effect Thruster that uses water vapour as the propellant. The aim of this project is to assess the potential performance of these thrusters at high powers, up to 20 kW. The performance analysis and optimization of the thruster will be conducted using a Particle-in-Cell (PIC) simulation based on an existing code called PlasmaSim, developed in C++, that will be modified to study this novel thruster design. No prior plasma physics experience is necessary, but I'm looking for students with an openness to learn new things. Through this project you will learn practical theory related to plasma thrusters, so it's an ideal opportunity for those interested in becoming involved in the field of spacecraft electric propulsion.

Project no:	IC-ACMAK07
Project title:	Miniaturized Hall Effect Thrusters fabricated using MEMS manufacturing techniques
Supervisor:	Knoll, Aaron Dr
Category:	Computational
Software:	PlasmaSim (in-house Particle-in-Cell plasma simulation in C++) FEMM Autodesk Fusion 360 Matlab
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Hall Effect Thrusters are one of the leading technologies used for satellite electric propulsion today. The purpose of this project is to investigate the feasibility of scaling these devices to ultra-miniature sizes, achievable using MEMS manufacturing methods. The exciting possibility is that the inherent increase in power density achievable at a miniature scale will allow these thrusters to function at much higher levels of background pressure – potentially sea level atmospheric conditions needed for terrestrial applications (e.g. miniature drones, aircraft, etc.). However, there is a risk that the intensity of the required magnetic field may be too high for practical applications. Additionally, the increased surface area to volume ratio is likely to constrain the efficiency. The purpose of this project is to evaluate the concept feasibility using PlasmaSim, a Particle-in-Cell (PIC) code that was developed in-house within the Aeronautics department (C++ language). Through this project you will learn practical theory related to plasma thrusters, so it's an ideal opportunity for those interested in becoming involved in the field of spacecraft electric propulsion.

Project no:	IC-AAEAK08/E
Project title:	Experimental testing of an E×B probe for plasma thruster plume characterization
Supervisor:	Knoll, Aaron Dr
Category:	Experimental
Software:	NI LabVIEW
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

Hollow cathode neutralizers are an essential component of almost every electric propulsion (plasma thruster) technology on the market. The purpose of the hollow cathode is to neutralize the outgoing ion beam to prevent charge accumulation on the satellite and, in the case of Hall Effect Thrusters, to provide the electron current to sustain the discharge. Despite their relative importance, these devices continue to be the 'weak link' in terms of the reliability and cost of the overall system. Additionally, there are virtually no hollow cathodes on the market that meet the needs of the growing small satellite industry.

This project builds on a design that was developed and constructed during a MSc project in 2020/21. The purpose of this current project is to demonstrate its successful operation within the large vacuum chamber facility (Boltzmann Chamber) and to characterize its

performance. No prior plasma physics experience is necessary, but I'm looking for students with an openness to learn new things. Through this project you will learn many practical skills that are specific to plasma thrusters, so it's an ideal opportunity for those interested in becoming involved in the field of spacecraft electric propulsion.

Kovac, Mirko Prof

Project no:	IC-MSCMK01/E
Project title:	Electronic design of gliding sensor controller
Supervisor:	Kovac, Mirko Prof
Category:	Experimental
Software:	
Confidential:	
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

Self-deploying sensors have great potential for precise automatic sensor delivery at long range for locations where the approach by an UAV is not possible.

This project will require the student to integrate an environmental monitoring IoT sensor with avionics components for a deployment - data acquisition mission.

Other work packages that can be developed include the integration of energy harvesting capabilities, mission extension and end-of-life management of the sensors.

Project no:	IC-MSCMK02/E
Project title:	Manufacturing of morphing structures
Supervisor:	Kovac, Mirko Prof
Category:	Experimental;#Experimental: Structures;#Experimental: Ready-made experiment;#Computational;#Theoretical;#Design
Software:	
Confidential:	
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

Morphing structures extend the operational capabilities of robots by allowing it's adaptation to different locomotion modes or to different environments. However, this structures need to be tailored to a specific application and require complex manufacturing methods.

This project will lead to the development of a workflow for the manufacturing of such structures. Some of work packages that can be developed are the following:

- (i) Address the integration of electroactive materials into the curing of a CFRP structure.
- (ii) Developing a strategy to manage heat dissipation
- (iii) Localized pre-curing of silicone impregnated fibres for stress release and morphing control.

Project no:	IC-MSCMK03/E
Project title:	Autonomous field MAV for aerial 3D printing
Supervisor:	Kovac, Mirko Prof
Category:	Experimental
Software:	
Confidential:	
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

Aerial 3D printing requires high accuracy state estimation and control. Currently the VICON motion capture system, which can only be used indoors, and model predictive controller are used to achieve the high accuracy indoor flight. To move the aerial 3D printing to outdoor environment, the student need to investigate how to enable the high accuracy state estimation without the help of VICON system, and modify the MPC controller to make it suitable for outdoor flight.

Project no:	IC-AAEMK04/E
Project title:	Modeling of morphing structures
Supervisor:	Kovac, Mirko Prof
Category:	Experimental;
Software:	
Confidential:	
Offered to:	MSc in Advanced Aeronautical Engineering

Morphing structures extend the operational capabilities of robots by allowing it's adaptation to different locomotion modes or to different environments.

The modeling of such structures and prediction of their deformation is how- ever challenging an there are different approaches worth exploring. Some of work packages that can be developed are the following: (i) Estimation of actuation requirements and prediction of structural failure given the Gaussian intrinsic curvature of the desired shape (ii) Identification of constitutive models for different electroactive materials (iii) Detailed modeling of the structure using FEM.

Laizet, Sylvain Prof

Project no:	IC-ACMSL01
Project title:	Simulation of a pulsating flow past a stenosed 2D artery with atherosclerosis
Supervisor:	Laizet, Sylvain Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Incompact3d and Paraview
Confidential:	
Offered to:	
	MSc in Advanced Computational Methods;

Atherosclerotic plaque can cause severe stenosis in the artery lumen. Blood flow through a substantially narrowed artery may have different flow characteristics and produce different forces acting on the plaque surface and artery wall. The disturbed flow and force fields in the lumen may have serious implications on vascular endothelial cells, smooth muscle cells, and circulating blood cells. In this work a simplified model is used to simulate a pulsating blood flow past a stenosed artery caused by atherosclerotic plaques of different severity. The focus is on a systematic parameter study of the effects of plaque size/geometry, flow Reynolds number, shear-rate dependent viscosity and flow pulsatility on the fluid wall shear stress and its gradient, fluid wall normal stress, and flow shear rate. The computational results obtained from this idealized model may shed light on the flow and force characteristics of more realistic blood flow through an atherosclerotic vessel. The Navier-Stokes equations and transport equation will be solved using the high-order flow solver Incompact3d (www.incompact3d.com). It is a Fortran 90 solver based on a Cartesian mesh, sixth-order finite-difference schemes and a fully spectral solver for the Poisson equation. The modelling of the artery is achieved thanks to an Immersed Boundary Method (IBM) based on a direct forcing approach to ensure the no-slip boundary condition at the wall of the artery. Previous experience using Fortran would be very useful but is not mandatory.

Project no:	IC-ACMSL02
Project title:	2D compressible flow solver for GPU hardware
Supervisor:	Laizet, Sylvain Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Paraview
Confidential:	
Offered to:	
	MSc in Advanced Computational Methods;

This project consists in writing from scratch flow solvers designed for GPU hardware. The solvers will deal with the 2D compressible Navier-Stokes equations and will be based on finite-difference methods on a Cartesian mesh. The aim is to investigate the robustness and performance of OpenMP, OpenACC and CUDA as parallel paradigms. Comparisons will be made with an equivalent Fortran solver dedicated to CPU hardware. The flow configuration of interest is the wake generated by a circular cylinder for low Reynolds numbers. Previous experience in GPU programming would be extremely useful but is not mandatory.

Project no:	IC-MSCSL03
Project title:	Computational Fluid Dynamics with augmented reality
Supervisor:	Laizet, Sylvain Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Paraview
Confidential:	
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

The main idea behind this project is to investigate the possibility of using augmented reality in computational fluid dynamics dataset to help conveying important information and results in a more efficient and rather intuitive way. The behaviour of fluid flow is often a complex paradigm for visualisation and cognitive interpretation. Mixed reality related technology can be a solution for enhanced virtual interactive learning environment. However, there are limited augmented reality platforms on fluid flow interactive learning. The project is aiming at developing an interactive education application for students to interact and understand the complexity of fluid flows, with a focus on the turbulence generated by wind turbines.

Project no:	IC-ACMSL04
Project title:	Real-time fluid simulations with Julia
Supervisor:	Laizet, Sylvain Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Julia
Confidential:	
Offered to:	
	MSc in Advanced Computational Methods;

This project consists in writing from scratch a flow solver in Julia which will allow for real-time visualisations. The solver will deal with the 2D compressible Navier-Stokes equations on a Cartesian mesh and will use an immersed boundary method to deal with fluid-structure interactions. Visualisations will be performed in real time via Pluto, a simple reactive notebooks for Julia. The flow configuration of interest is the wake generated by a circular cylinder for low Reynolds numbers. Previous experience in Julia would be extremely useful but is not mandatory.

Project no:	IC-MSCSL05
Project title:	Prediction models for gravity currents
Supervisor:	Laizet, Sylvain Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Matlab or Python
Confidential:	
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Gravity currents are common in nature, either in the atmosphere due to sea-breeze fronts, in avalanches of airborne snow or debris flows, or in the ocean due to turbidity currents and river plumes. They are created by the release of a fixed volume of a heavy suspension into a lighter ambient fluid. The aim of this project is to investigate various models to predict the extent of gravity current in various configurations. The gravity current will be modelled by either the single- or the two-layer shallow-water equations, the two-layer equations being necessary to include the effects of the overlying fluid, which are important when the depth of the current is comparable to the depth of the overlying fluid. Comparison will be made with existing experimental and numerical data. Previous experience using Matlab or Python would be very useful but is not mandatory.

Project no:	IC-ACMSL06
Project title:	Fortran or C for Computational Fluid Dynamics: which one is the fastest on CPUs?
Supervisor:	Laizet, Sylvain Prof
Category:	Computational;#Numerical
Software:	
Confidential:	
Offered to:	
	MSc in Advanced Computational Methods

This project consists in writing from scratch two flow solvers, one in Fortran and one in C. The solvers will deal with the compressible Navier-Stokes equations. The solvers will be based on finite-difference methods on a Cartesian mesh. Once the solvers are up and running, comparison will be made with various compilers and various CPUs to evaluate which one is the fastest. The flow configuration of interest is the wake generated by a circular cylinder for low Reynolds numbers. Previous experience in Fortran and/or C would be extremely useful but is not mandatory.

Project no:	IC-MSCSL07
Project title:	Fluid simulations with the Domain Specific Library OPS
Supervisor:	Laizet, Sylvain Prof
Category:	Computational;#Theoretical;#Numerical
Software:	OPS library
Confidential:	
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project consists in writing from scratch an incompressible flow solver using the Domain Specific Library OPS. OPS (Oxford Parallel library for Structured mesh solvers) is a high-level embedded domain specific language for writing multi-block structured mesh algorithms, and the corresponding software library and code translation tools to enable automatic parallelisation of the high-level code on multi-core and many-core architectures. The aim of the project is to leverage OPS to automatically write an incompressible flow solver suitable for modern hardware (CPUs/GPUs). The solver will be based on high-order implicit finite-

difference schemes on a Cartesian mesh. The reference test case will be the Taylor Green Flow, a well-known benchmark in the Computational Fluid Dynamics community which will be used for performance analysis. Comparisons will be made with a FORTRAN flow solver on multiple CPUs and GPUs.

Project no:	IC-MSCSL08
Project title:	Quantum algorithms to solver PDEs
Supervisor:	Laizet, Sylvain Prof
Category:	Computational;#Theoretical;#Numerical
Software:	
Confidential:	
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

With the advent of quantum computing (QC), a plethora of quantum algorithms have demonstrated, both theoretically and experimentally, more powerful computational possibilities than their classical counterparts. The project will investigate methods to solve linear and non-linear PDEs. The project will be using the Quantum Exact Simulation Toolkit, which is a high performance simulator of universal quantum circuits, state vectors and density matrices. QuEST can be used by both C and C++, is open source, multithreaded, distributed and GPU-accelerated. Needing only compilation, QuEST is easy to run both on laptops and supercomputers, where it can take advantage of multicore and networked machines to quickly simulate circuits on many qubits.

Lee, Koon-Yang Prof

Project no:	IC-COMKYL01/E
Project title:	Upgrading waste: Design and manufacturing of novel polymer composites reinforced with waste materials
Supervisor:	Prof Koon-Yang Lee
Category:	Experimental;#Computational;#Design
Software:	Abaqus / 3DX
Confidential:	No
Offered to:	
	MSc in Composites

Consumer plastics are highly-engineered products with well-defined physical properties in order to meet their stringent design specification, which includes the use of additives and multi-polymer products. These must be separated into recognisable plastic grades prior to recycling. Melt reprocessing of mixed plastic waste leads to products with inferior properties due to incompatibility between the different plastics; these are instead down-cycled as fillers for building materials. To bridge this property-performance gap, we propose a composite approach combining mixed plastic waste with waste material to produce high performance engineering materials, upgrading the value of mixed plastics for advanced engineering applications. In this project, the student(s) will study the structure-property relationship of

mixed plastic waste (if applicable) and identify novel suitable processing/reinforcement, as well as its architecture that will upgrade waste plastics. The grand vision of this project is to develop material solutions that will contribute to a more sustainable future.

Up to two positions are available for this project. Students are encouraged to contact Dr Lee for more information to discuss workload and research directions.

Project no:	IC-COMKYL02/E
Project title:	Optically transparent material derived from cellulose nanofibres
Supervisor:	Prof Koon-Yang Lee
Category:	Experimental
Software:	
Confidential:	Yes
Offered to:	
	MSc in Composites

Bacterial cellulose (BC) is synthesised as a tough and continuous nanofibre network with high specific surface area, with single fibre tensile modulus and strength of up to 160 GPa and 6 GPa, respectively. We hypothesise that this continuous BC nanofibre network will serve as excellent nano-reinforcement for polymers with similar refractive indices, such as acrylic resin, for the production lightweight polymeric transparent armour with specific energy absorption higher than the state-of-the-art (e.g. impact-modified acrylic and polycarbonate-impact-modified acrylic resin laminates), thereby bridging the gap between monolithic polymer and glass-clad polycarbonate. The introduction of such continuous nanofibre network into acrylic resin will introduce additional energy-absorbing mechanisms, including fibre de-bonding (due to an increase in fibre-polymer matrix interface), fibre re-orientation and fracture, enhancing the impact resistant of the resulting BC-reinforced acrylic resin without sacrificing optical transparency (due to its refractive index matching with acrylic resin and small length scale of 50 nm). This proposed research aims to interface BC engineering and polymer sciences with materials engineering and processing. Through this interface, this project will explore challenging and high-impact methods for the development of new large-scale processes in order to manipulate the microstructure of BC and efficiently utilise them as nano-reinforcement for novel lightweight transparent armour.

One position is available for this project and potential students are encouraged to contact Dr Lee for more information to discuss workload and research direction

Project no:	IC-COMKYL03/E
Project title:	Upgrading individual protective clothing with nanocellulose – Project 1
Supervisor:	Prof Koon-Yang Lee
Category:	Experimental
Software:	Abaqus / 3DX
Confidential:	Yes
Offered to:	
	MSc in Composites

Individual protective equipment (IPE) is used to protect personnel from the effects of chemical, biological and radiological (CBR) warfare agents. The technology used in IPE clothing provides a very high level of protection against CBR warfare agents for the military personnel. However, the physiological burden of wearing such clothing can be high in some circumstances. One major challenge is the removal of body heat, which is facilitated by the use of air-permeable fabrics as these allow water vapour derived from sweat to escape the microclimate of the suit. However, if the air-permeability is too high the levels of protection can fall significantly. Consequently, there is an interest in developing protective technologies that could provide protection without impeding the permeation of water vapour out of the suit. It is anticipated that the application of nanocellulose in IPE clothing will foster the development of lighter, multifunctional IPE garment with reduced physiological burden to the user by providing filtration within the outer shell textile. As nanocellulose is hydrophilic in nature it is anticipated that this will also facilitate the passage of moisture from the inner layers to the exterior of the garment, allowing for efficient body heat removal. Therefore, the aim of this proposed project is to produce nanocellulose-enhanced IPE and to demonstrate the scale of the resulting performance improvements over the state-of-the-art. It should be noted that no chemical, biological and radiological agents will be used in our experiments.

Up to two positions are available for this project and potential students are encouraged to contact Dr Lee for more information to discuss workload and research directions.

Project no:	IC-COMKYL04/E
Project title:	Upgrading individual protective clothing with nanocellulose – Project 2
Supervisor:	Prof Koon-Yang Lee
Category:	Experimental
Software:	Abaqus / 3DX
Confidential:	Yes
Offered to:	
	MSc in Composites

See project description above.

Project no:	IC-COMKYL05/E
Project title:	Manufacturing "artificial wood" from low cost ionic liquids
Supervisor:	Prof Koon-Yang Lee
Category:	Experimental
Software:	
Confidential:	Yes
Offered to:	
	MSc in Composites

There is currently a timely need to design and manufacture renewable materials for high volume structural applications and decouple our economy from fossil-derived non-renewable resources. Cellulose-based natural fibres are the prime candidate for the production of low-cost high-performance renewable composites. However, there is still a property-performance gap between natural fibre-reinforced polymers and traditional fossil-derived engineering materials, as the high tensile stiffness (up to 165 GPa) and strength (at least 1 GPa) of cellulose microfibrils have yet to be fully exploited in a composite setting. Nature has been

very efficient at manipulating and exploiting cellulose microfibrils in wood (a natural composite) to produce high performance materials. This project will take inspiration from wood and manufacture the world's first "artificial wood", i.e. cellulose microfibril-reinforced lignin composites with the native cellulose-I structure preserved (mimicking wood cell wall), using simple and intrinsically scalable manufacturing concepts. The proposed research activities are structured around (i) manufacturing "artificial wood" from (low cost) ionic liquid, and (ii) optimising the techno-economics and lifecycle of "artificial wood" manufacturing. It is envisaged that the resulting "artificial wood" will target engineering applications that cannot be achieved by conventional bio-based polymers or renewable natural fibre-reinforced polymers alone and could serve as alternative to traditional glass fibre-reinforced polymers. Up to two positions are available for this project and potential students are encouraged to contact Dr Lee for more information to discuss workload and research directions.

Levis, Errikos Dr

Project no:	IC-MSCEL01
Project title:	Design of Small UAVs for Joined Formation Flight
Supervisor:	Levis, Errikos Dr
Category:	Computational;#Design
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Current research into formation flying has concentrated on close or extended formation flying, which promises reductions of induced drag to a theoretical maximum of 50% for trailing aircraft. Joining multiple vehicles in flight would allow the wetted area and aspect ratio to be manipulated, optimising the swarm for the intended mission. For example joining the lifting surfaces of just two aircraft side by side could result in increases of up to 40% in endurance or 20% in range. Even greater improvements in range, and an increase in optimum cruise speed would be obtained if the wings of the aircraft were to overlap, reducing the overall wetted area and associated viscous drag and increasing the overall wing loading. Past work has shown substantial improvements in aerodynamic performance are possible but highlighted issues with vehicle stability and the added weight of joining mechanisms. This project will concentrate on the effect of different joining formations on the vehicle's overall mission performance.

Project no:	IC-AAEEL02
Project title:	Hybrid-Electric distributed propulsion for aircraft
Supervisor:	Levis, Errikos Dr
Category:	Computational;#Design
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

In June 2016, NASA announced that the X-57 research plane would be used to study the aerodynamic and propulsive efficiency improvements made possible by the use of hybrid electric distributed propulsion systems. Projects on this area will build on the work of past students and concentrate on:

- 1) Investigating the applicability of such technologies to aircraft of different types and sizes.
- 2) The design and optimization of a wing-distributed propulsion system.
- 3) The effects that using such systems may have on the stability and controllability of aircraft.

A good understanding of the principles of aircraft design and good coding skills are required.

Project no:	IC-MSCEL03
Project title:	Prediction of porpoising onset for seaplane hulls
Supervisor:	Levis, Errikos Dr
Category:	Computational;#Theoretical
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

The issue of porpoising is one encountered by speedboats and seaplanes when traveling at high speed over water and can have catastrophic consequences. Porpoising is a dynamic pitch-heave instability caused by the effect of a hull's angle of attack and draught on the hydrodynamic lift and pitching moments. This project will concentrate on predicting the angle of attack at which porpoising is likely to become an issue at varying angles of attack. A literature review of available methods will be carried out and the most promising ones will be implemented in order to identify the porpoising limits of any arbitrary seaplane hull. The effects of a seaplane's aerodynamic properties on porpoising will be investigated and results will be compared with experiments.

Project no:	IC-MSCEL04
Project title:	Flight Mechanics and Control of UAVs in joined flight using hinged connections
Supervisor:	Levis, Errikos Dr
Category:	Computational;#Design; Analysis
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Past work has shown that joining several aircraft tip-to-tip can offer significant aerodynamics improvements. The increased resulting wing aspect ratio however significantly increases the bending moments acting on the inboard aircraft's wings, often resulting in a significant weight penalty if more than 3-4 aircraft are joined together. One method by which bending

moments and therefore the associated structural weight penalty could be reduced, is the use of hinges to connect the aircraft. This project aims to investigate the impact of introducing such a coupling method on the aircraft's handling qualities. The project will involve deriving the equations of motion for the joined aircraft configuration, development of an appropriate unsteady aerodynamic model to approximate the stability derivatives and an analysis of the aircraft's response to perturbations and control inputs. This project is well suited for a student with strong coding skills and an interest in flight mechanics and unsteady aerodynamics.

Project no:	IC-MSCEL05
Project title:	Design and Optimization of Wing-In-Ground Effect vehicles
Supervisor:	Levis, Errikos Dr
Category:	Computational;#Design
Software:	matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

The Wing-In-Ground Effect (WIG) concept has been around for some time and was investigated in depth by Soviet designers, resulting in number of vehicles known as Ekranoplans. Due to the potential reduction in drag achieved by flying in close proximity to the ground the concept has recently been revived by Boeing and other research institutes. Flying near the surface however does also complicate the aerodynamics and stability of the craft. The project will build on the work of previous students to design and optimize sample aircraft and evaluate the effect that size and operating points have on their performance and thus commercial viability.

Project no:	IC-AAEEL06
Project title:	Improved Weight and Power Requirement Estimation for Aircraft Subsystems
Supervisor:	Levis, Errikos Dr
Category:	Literature;#Design; Numerical
Software:	matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

The weight estimation of aircraft subsystems in early design stages has to date largely relied on empirical relations derived from past designs of such subsystems. Such methods however provide little insight into the impact of subsystem reliability and performance requirements on the system weight, and electrical and mechanical power requirements. In

order for the benefits of novel ideas such as more electric and all electric aircraft to be investigated, more detailed sizing methods for key aircraft subsystems must be developed. For example, if the control system of the aircraft is electrified, will the reduced weight of the hydraulic system and the reduced mechanical power extracted from the power plant be greater than the weight penalty associated with additional wiring, actuators and fuel for electrical power generation? The aim of this project is therefore to design a parametric subsystem design tool for an aircraft fuel, hydraulic and/or electrical system,, capable of predicting subsystem weight and power requirements subject to a set of performance and reliability requirements.

Li, Qianqian Dr

Project no:	IC-MSCQL01/E
Project title:	Advanced materials for hypersonic vehicles
Supervisor:	Dr Qianqian Li (with Dr Dr Georgios Rigas and Dr Paul Bruce)
Category:	Experimental;#Computational;#Theoretical;#Design;#Analysis;#Literature
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

While the aerodynamic challenges related to hypersonic aerospace vehicles have been frequently explored in the past, the advanced materials which could be suitable for providing the desired properties in these applications have not been well-explored yet. Targeting certain properties such as heat flux and heat conductivity of aerospace components for given surface boundary conditions, potential materials and composites, together with their appropriate manufacturing methods and structural design, will be investigated in this project. The project student will first model and calculate the required properties for components of a representative aerospace vehicle at hypersonic conditions, and then match possible materials/composites and manufacturing ideas to the selection. Particularly nano-materials and nanocomposites will be focused on as they provide the possibility of high specific strength, multifunctionality and tuneability of properties. This multidisciplinary project spans the fields of high speed aerodynamics and advanced materials and offers a unique opportunity to explore the research challenges in these areas.

Project no:	IC-MSCQL02/E
Project title:	Damage detection in structure of nanocarbon lightweight metal composite
Supervisor:	Dr Qianqian Li (co-supervisor Dr Zahra Sharif Khodaei)
Category:	Experimental;Structures;#Theoretical;#Analysis;#Literature
Software:	Matlab
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

Nanocarbon (carbon nanotubes and graphene) reinforced lightweight metal composites have attracted considerable attention due to their excellent performance as super-strong lightweight structural materials with high electrical and thermal conductivities. These materials have great potential for aeronautic and automotive applications. In order to understand its industrial potential for real application, this project is aiming to investigate the applicability of Structural Health Monitoring (SHM) techniques on such nanocarbon lightweight weight composites. The SHM technique to be investigated is based on an array of Piezoelectric (PZT) transducers, actuating Lamb waves in the structure and sensing the waves after propagating. The changes in the propagation properties due to damage can be used for detecting and characterising the damage. There are various algorithms developed for detecting and characterising damage based on comparing the propagational properties of the wave in pristine and damaged state.

Project no:	IC-MSCQL03/E
Project title:	Carbon nanotube reinforced lightweight metal composites via melt stirring
Supervisor:	Dr Qianqian Li
Category:	Experimental;#Experimental: Manufacturing intensive;#Experimental: Ready-made experiment;#Theoretical;#Analysis;#Literature
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

Carbon nanotubes have attracted great attention since their discovery due to their excellent physical and mechanical properties. They are considered as ideal reinforcements to produce composites. Much research has been done on CNT/polymer composites; while CNT/metal composites have not been fully exploited due to the more difficult production process and more complicated reinforcing mechanisms, despite high potential of such composites. In this project, the student will produce CNT reinforced magnesium composites via melting route and investigate the composites mechanical properties and microstructure. The production parameters will be investigated and optimised.

Project no:	IC-MSCQL04/E
Project title:	Nanocarbon (carbon nanotubes and graphene) based polymer composites by 3D printing
Supervisor:	Dr Qianqian Li (co-supervisor Dr Vito Tagarielli)
Category:	Experimental;#Experimental: Structures;#Experimental: Manufacturing intensive;#Experimental: Ready-made experiment;#Design;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

Nanocarbon such as carbon nanotubes and graphene based polymer composites have become a hot topic in composites in recent years. This is due to nanocarbon extraordinary physical and mechanical properties (very high Young's modulus, high electrical conductivity etc.), which makes it a highly interesting candidate for multifunctional composites. In this project, the student will optimise the production process and check the influence of the different parameters on the procedure (e.g. different concentration of graphene, different dispersion methods...) of the filaments. Manufacture of such composites using 3D printing will also be carried out. The characterisation of the mechanical properties and microstructure of the composites will be another focus point.

Lubbock, Roderick Dr

Project no:	IC-MSCL01
Project title:	Design optimisation of sports helmets
Supervisor:	Dr Roderick Lubbock
Category:	Computational;#Design;#Analysis
Software:	3D Experience SIMULIA
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

There are numerous designs of sports helmets specific to a number of sports, however these all have more or less the same purpose: to prevent harmful brain decelerations. The aim of the proposed project is therefore to use design optimisation to develop an optimum sports helmet design to meet numerous helmet testing standards that could therefore find application in multiple sports.

Extensions of the project could look at the use of additive manufacturing to generate optimised designs. Such a design could incorporate 3D scanning to develop cost-effective helmets custom made to fit individual users, that could bring offer further benefits in reducing harmful decelerations.

Students interested in this project must have an excellent understanding of structural mechanics, as well as the underlying mathematics of design optimisation (e.g. topology optimisation, generative design) and be confident using industry standard FEA software such as 3DExperience SIMULIA (Abaqus FEA).

Project no:	IC-AAERL02/E
Project title:	A spatial impulse response technique for improving the frequency response of thermocouple probes
Supervisor:	Roderick Lubbock
Category:	Experimental;#Theoretical;#Design;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

Thermocouple probes are ubiquitous in the field flow temperature measurement however they are limited by their low frequency response due to the large thermal inertia of the bead. This experimental project looks at improving the probe frequency response for a given flow by deriving the probe impulse response by traversing it through a spatially varying temperature field at different known velocities. This technique has already been shown to work for pressure probes [1].

Experimental aspects of the project and will involve building a simple experimental rig for traversing thermocouple probes across a hot flow (e.g. from a heat gun) at different velocities, along with a simple PC-based DAQ system. Theoretical aspects will include developing an excellent understanding of thermocouple-probe based flow temperature measurements, whilst analytical aspects will include time-frequency analysis (impulse response, deconvolution, FFTs etc.).

[1] Burdett, Lubbock & Povey, 2018. AN IMPULSE RESPONSE TECHNIQUE TO IMPROVE THE EFFECTIVE FREQUENCY RESPONSE OF PRESSURE PROBES. XXIV Biennial Symposium on Measuring Techniques in Turbomachinery, Prague, Czech Republic, 29 - 31 August 2018. Paper No MTT2418B23. Available from:

https://meastechturbo.com/paper-archives/2418-prague-2018/item/download/466_1bdd5c50c7ece7b207acee25244b0fac

Project no:	IC-AAERL03/E
Project title:	A cycling/sports helmet with air filtration – experimental
Supervisor:	Roderick Lubbock
Category:	Experimental;#Experimental: Manufacturing intensive;#Design
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

FFP2 respirator face masks have several problems:

- They are uncomfortable to breath through under conditions of high exertion, for instance when cycling or running.
- They typically leak around the nose area making them incompatible with sunglasses, which suffer from condensation under heavy breathing.
- They trap moisture against the skin and can be irritating to wear for longer periods.
- They are inconvenient to put on around the ears whilst wearing a helmet, e.g. for cycling or sport.

The aim of this project is therefore to design, build and test a simple proof of concept experimental prototype of a practical cycling/sports helmet that is able to filter out harmful air particles (ideally to FFP2 standard) without causing breathing discomfort or affecting the wearers vision.

Students interested in this project must have an excellent understanding of undergraduate fluid mechanics and thermodynamics and must be confident designing and building prototypes and carrying out independent laboratory experiments.

Project no:	IC-MSCL04/E
Project title:	Design of an assistive knee brace – experimental
Supervisor:	Roderick Lubbock
Category:	Experimental;#Experimental: Structures;#Experimental: Manufacturing intensive
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering
	MSc in Composites

Knee problems such as osteoarthritis and sports injuries to cartilage and ligaments affect millions of people around the world, and can have severe impacts on patients' physical and mental health. Assistive knee braces allow these patients to live a more normal active lifestyle. Numerous knee brace designs have been proposed [1] however the design problem is far from solved. This project looks at application of modern design methods and manufacturing techniques to the design and potentially testing of a new assistive knee brace.

Students interested in this project must have an excellent understanding of structural mechanics and mechanical design and must be confident in designing, building and testing prototypes and carrying out independent laboratory experiments.

[1] L. Zhang, G. Liu, B. Han, Z. Wang, H. Li and Y. Jiao, "Assistive devices of human knee joint: A review", Robotics and Autonomous Systems, vol. 125, p. 103394, 2020. Available: 10.1016/j.robot.2019.103394.

Project no:	IC-MSCL05
Project title:	Design optimisation of knee braces
Supervisor:	Roderick Lubbock
Category:	Computational;#Design;#Analysis
Software:	3D Experience SIMULIA
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering
	MSc in Advanced Computational Methods
	MSc in Composites

There are numerous knee braces on the market featuring a variety of different designs. There appears to have been little work done on design optimisation of supportive knee braces. Many of the braces on offer are also very expensive.

The aim of the proposed project is therefore to use design optimisation to develop an optimum low-cost knee brace design. Extensions of the project could look at the use of additive manufacturing to generate further optimised designs. Such a design could incorporate 3D scanning to develop cost-effective braces custom made to fit individual users. Currently such braces are available but are prohibitively expensive.

Students interested in this project must have an excellent understanding of structural mechanics, as well as the underlying mathematics of design optimisation (e.g. topology optimisation, generative design) and be confident using industry standard FEA software such as 3DExperience SIMULIA (Abaqus FEA).

Project no:	IC-MSCL06
Project title:	Integrated helmet - breathing apparatus for avalanche survival - computational
Supervisor:	Roderick Lubbock
Category:	Computational;#Theoretical;#Design
Software:	3D Experience SIMULIA
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering
	MSc in Advanced Computational Methods

85% of avalanche deaths are caused by asphyxiation due to the inability of the victim to evacuate exhaled CO₂ from around their airways due to burial in snow. Almost two decades ago [1] a winter sports manufacturer developed a solution to this called the "Avalung", which was essentially a breathing tube for drawing in fresh air (through the snow) from a point behind the victim away from their airways, meaning that the victim is able to avoid rebreathing their own CO₂. The manufacturer claimed their device would increase the avalanche burial survival window from 15 minutes to 58 minutes (although this does not appear to have been supported by independent test data).

The Avalung had a major design flaw however (e.g. [2], [3]), which was the requirement of the victim to either insert the breathing tube in their mouth once they were already in the avalanche - a very difficult thing to do while being violently thrown down a mountain - or to ski with it in their mouth leading to obvious discomfort. The product appears to have been discontinued and is no longer for sale on the manufacturers website at the time of writing. Aside from these impracticalities the basic principle of the Avalung - evacuating CO₂ away from the user's airways - appears to be sound. This project seeks to overcome the impracticalities of the Avalung by integrating some sort of avalanche breathing apparatus into a helmet. Helmets are a sensible platform for a breathing apparatus since their use is becoming more and more common amongst snow sport enthusiasts, with many ski resorts and snow sports competitions now making them compulsory.

The challenges associated with this project include modelling air flow under conditions of snow burial, as well as modelling air flows through the device itself.

Students interested in this project must have an excellent understanding of fluid mechanics, as well as the underlying mathematics of computational fluid dynamics and be confident using industry standard CFD software such as 3DExperience SIMULIA and ANSYS FLUENT

[1] Dawson, L., BLACK DIAMOND AVALUNG – REVIEW, 30th September 2004, <https://www.wildsnow.com/13925/black-diamond-avalung-review/>, [accessed 1st May 2022].

[2] NEWSCHOOLERS Forums, Avalung discontinued?, <https://www.newschoolers.com/forum/thread/888877/Avalung-discontinued>, [accessed 1st May 2022].

[3] Brown, J., Should You Use An Avalung?, 7th February 2018, <https://www.powder.com/gear-locker/should-you-use-an-avalung/>, [accessed 1st May 2022].

Magri, Luca Dr

Project no:	IC-ACMLM01
Project title:	Quantum algorithms for data-driven fluid mechanics
Supervisor:	Magri, Luca Dr
Category:	Computational
Software:	Python
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

The student will implement quantum algorithms to either (i) train physics-aware neural networks to learn the chaotic dynamics of fluids, or (ii) simulate the dynamics of fluid mechanics systems (to be agreed with the supervisor). The fluid mechanics systems will be fundamental, chaotic and reduced-order models of real flows. The main objective is algorithmic and methodological, rather than physical. The goal is to show the quantum advantage (polynomial/exponential acceleration of computations) on quantum simulators for a series of elementary dynamical systems.

Montomoli, Francesco Prof

Project no:	IC-MSCFM01
Project title:	Non Intrusive Polynomial Chaos for transonic/supersonic flows
Supervisor:	Montomoli, Francesco Prof
Category:	Computational
Software:	Matlab, OpenFoam
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

The aim of the project is to develop non intrusive polynomial chaos for transonic flows based on stochastic collocation methods. Uncertainty plays a key role in many modern applications and it is important to have a reliable method to account such uncertainty. A stochastic collocation methodology with Pade's polynomials will be used to simulate the stochastic field. This project requires a high level of mathematical ability and experience using Matlab.

Project no:	IC-MSCFM02
Project title:	Optimization of combustor mixer
Supervisor:	Montomoli, Francesco Prof
Category:	Computational, Design
Software:	Matlab, OpenFoam
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

The aim of this project is to study and optimize a realistic model of gas turbine component.

Project no:	IC-MSCFM03
Project title:	AI for Design for Additive Manufacturing; free support design
Supervisor:	Montomoli, Francesco Dr
Category:	Computational
Software:	Matlab, OpenFoam
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;
	MSc in Composites

The aim of the project is to implement techniques of dfAM using free support strategies.

Project no:	IC-MSCFM04
Project title:	Uncertainty Quantification and Space: collaboration with NASA
Supervisor:	Montomoli, Francesco Dr
Category:	Computational;#Theoretical;#Numerical
Software:	Matlab, in house codes
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

The project will investigate advanced UQ models for space components and launchers in collaboration with NASA Langley

For this project we need a very good student with a deep understanding of mathematics and/or aeronautics

Project no:	IC-MSCFM05
Project title:	Uncertainty Quantification and alien species
Supervisor:	Montomoli, Francesco Dr
Category:	Computational
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

Alien dispersion on complex areas is a challenging task due to the unknown parameters. For this reason this project will analyse the alien dispersion in a realistic environment considering the impact of unknown parameters.

Morrison, Jonathan Prof

Project no:	IC-MS CJM01/E
Project title:	Instrumentation for CRM-HL testing – Project 1
Supervisor:	Morrison, Jonathan Prof (co-supervisor Dr Kevin Gouder)
Category:	Experimental
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

A Boeing generated project based on the open-source Common Research Model - High Lift programme. Details will be determined by QinetiQ who test the CRM model in the 5m tunnel at Farnborough. A half-span model is due to come to the 10x5 tunnel next year.

Project no:	IC-MS CJM02/E
Project title:	Instrumentation for CRM-HL testing – Project 2
Supervisor:	Morrison, Jonathan Prof (co-supervisor Dr Kevin Gouder)
Category:	Experimental
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

See project description above.

Project no:	IC-MSJCM03/E
Project title:	Ribbon-induced transition in a Blasius boundary layer
Supervisor:	Morrison, Jonathan Prof
Category:	Experimental
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

We have previously used an oscillating ribbon above a boundary layer to induce transition in a laminar boundary layer. The 2D ribbon oscillates vertically above the wall generating the boundary layer. At certain frequencies, TS waves are produced through the process of receptivity (which can be both linear and, at larger amplitudes, nonlinear) where roughness-induced disturbances at the surface resonate with the longer wavelengths generated by the ribbon. At large ribbon amplitudes, modal transition is bypassed, so that Klebanoff-like disturbances are produced. We wish to know why: this experimental study to be performed in T1 will be supplemented by stability code simulations. We would also like to develop the project using time-dependent surface roughness.

Project no:	IC-MSJCM04/E
Project title:	Dynamic roughness
Supervisor:	Morrison, Jonathan Prof
Category:	Experimental
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

We wish to design an experiment to test the use of dynamic roughness for the control of turbulent boundary layers. The dynamic roughness will be implemented through controlled actuation of chequerboard bifurcations in compliant membrane-substrate structures. This highly novel approach enables the simultaneous change in the time-dependent surface conditions so that the moving surface may be characterised by both an imposed lengthscale (wavenumbers) and a timescale (frequency), with the added motivation of identifying the drag-altering behaviour. Such an approach replicates many practical examples of time-dependent surface change used for boundary layer control in such as the delay of separation. The project would initially involve a design of the boundary layer experiment followed by a bench top experiment in, for example, fully developed turbulent channel flow. The project is intrinsically related to the project of same name supervised by Dr. Santer and focusing on the design and construction of the chequerboard.

Project no:	IC-MSCJM05/E
Project title:	Near-field PIV measurements of a forced axisymmetric wake
Supervisor:	Morrison, Jonathan Prof
Category:	Experimental
Software:	
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

The wake produced by a three-dimensional bluff body in uniform flow has always been a challenge for fundamental research in fluid mechanics. Even today, it is impossible to predict accurately the fluid force exerted on a bluff body because of the intricate combination of several phenomena: flow separation, local instabilities, global instabilities, recirculating flow and turbulence – irrespective of the body geometry. In the simplest case of a sphere, the difficulty in modelling the flow first appears in the laminar regime, in which the wake undergoes a succession of non-trivial bifurcations as the Reynolds number increases. At the first bifurcation, the steady laminar wake loses its axisymmetry. Then, at a second bifurcation, the wake becomes unsteady with periodic antisymmetric fluctuations, while on average, preserving the global asymmetry resulting from the first bifurcation. On the long-time average, the wake gradually loses the asymmetry introduced by the first bifurcation, and the axisymmetry of the mean flow is finally restored from $Re > 1000$. An initial experimental study in the T2 tunnel is to be conducted on a sphere and using the motion controller and drag balance.

Project no:	IC-MSCJM06/E
Project title:	Inertial subrange in turbulent pipe and channel flow - Project 1
Supervisor:	Morrison, Jonathan Prof
Category:	Experimental, Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

Using Princeton superpipe data, we have been looking at the behaviour of the inertial subrange in high-Reynolds-number turbulent pipe flow $Re_{\lambda} \sim 1000$. Even in a simple flow such as this, the energy transfer is not classical, as described by the $-5/3$ law in spectra, or the "4/5ths" law in real space using structure functions. The goal is to investigate the effects of small shear in the mean velocity profile by extending the analysis to small distance from the pipe centre line as well as in the local equilibrium region. The data are single axial velocity component from the NSTAP probe at Re_{τ} up to 98000. The second database involves DNS data from a turbulent channel flow at $Re_{\tau} = 5200$ which broadly has similar behaviour of the energy budget as that in pipe flow, http://journals.cambridge.org/article_S0022112015002682. Here we have the fully resolved 3D data and more ambitious approaches can be used, in particular using the Kármán-Howarth-Monin equation for low-order structure functions.

Project no:	IC-MSJCM07/E
Project title:	Inertial subrange in turbulent pipe and channel flow - Project 2
Supervisor:	Morrison, Jonathan Prof
Category:	Experimental, Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

See project description above.

Project no:	IC-MSJCM08/E
Project title:	Analysis of the effects of surface waves on turbulent boundary layers for drag reduction
Supervisor:	Morrison, Jonathan Prof
Category:	Experimental, Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods;

We have been developing kagome lattices for drag reduction and the analysis of the effect of surface waves using surface measurements of digital image correlation DIC and the near-velocity field using PIV. The basic ideas and the initial experiment is described in the paper, <https://doi.org/10.1007/s10494-018-9926-2>. Here we are developing a novel system in which green PIV light is used to measure fluid measurement simultaneously and distinct from measurement of the surface motion using DIC. The latter uses the speckle pattern from the surface, in which blue light fluoresces to produce a wavelength shift and a reflection of orange light. With the appropriate narrow-band filters, to minimise interference between the two signals, we can make examine the effect of the forcing on the boundary layer in detail.

Mylvaganam, Thulasi Dr

Project no:	IC-ACMTM01
Project title:	Control of autonomous vehicles in challenging environments
Supervisor:	Mylvaganam, Thulasi Dr
Category:	Computational;#Theoretical;#Analysis;#Numerical;#Literature
Software:	Matlab
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Airborne, ground and marine autonomous vehicles are rapidly gaining interest for a large variety of applications, in particular in the context of automated operation in unknown and challenging environments. Such applications rely heavily on the underlying control systems. For reliable and efficient operation it is essential that objectives such as “optimality” or “robustness” are taken into account in the control design. In this project the student will focus on marine autonomous surface vehicles (ASVs) and design control laws which ensure that one or more ASVs are manoeuvred in a desired manner taking into consideration that the manoeuvre should be done in an optimal and/or robust manner. The task can be achieved in a single step or in two steps (i.e. trajectory planning and trajectory tracking). The results will be demonstrated in simulations using Matlab.

The work will be of a highly mathematical nature (including elements of nonlinear systems and nonlinear control). It is therefore absolutely essential that the student has a strong background in mathematics, dynamical systems and control theory. Familiarity with using Matlab is also required.

Project no:	IC-ACMTM02
Project title:	Optimal control of marine vehicles
Supervisor:	Mylvaganam, Thulasi Dr
Category:	Computational;#Theoretical;#Analysis;#Survey;#Numerical;#Literature
Software:	MATLAB
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Airborne, ground and marine autonomous vehicles are rapidly gaining interest for a large variety of applications. Such applications rely heavily on the underlying control systems. For efficient operation it is essential to take into account "optimality" in the control design. In this project the student will focus on marine autonomous vehicles (surface or subsea) and design control laws which ensure that one or more such vehicles perform a desired task (e.g. achieving a certain desired motion, following a desired path, etc.) efficiently and reliably. Namely, the student will consider optimal control in the context of surface/subsea marine vehicles. The results will be demonstrated in simulations using Matlab.

The work will be of a highly mathematical nature (including elements of nonlinear systems and nonlinear control). It is therefore absolutely essential that the student has a strong background in mathematics, dynamical systems and control theory. Familiarity with using Matlab is also required.

Project no:	IC-ACMTM03
Project title:	Autonomous Monitoring
Supervisor:	Mylvaganam, Thulasi Dr
Category:	Computational;#Theoretical;#Analysis;#Numerical;#Literature
Software:	MATLAB
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

In this project, the student will focus on multi-joint robots (such as robotic arms or snake-like robots for ground, space or subsea operation). The student will formulate and solve control problems for the autonomous coordination and manoeuvring such robots to capture the overall aim of monitoring/covering a given region. The problem formulation could involve optimal control and/or game theory. The results will be demonstrated via simulations in Matlab. The formulation and solution of the coverage problem could, for instance, be of interest for robot-assisted response to a fire, autonomous monitoring (e.g. for subsea pipelines, space structures), etc.

The project will be of a highly mathematical nature, likely including elements of nonlinear control. Thus, a strong background in mathematics and control theory is absolutely essential. The project will involve programming in Matlab, so familiarity with using Matlab is required.

Project no:	IC-ACMTM04
Project title:	Optimal control of wind turbines
Supervisor:	Mylvaganam, Thulasi Dr
Category:	Computational;#Theoretical;#Numerical;#Literature
Software:	Matlab, NREL's OpenFAST (or other alternative simulation tools)
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

As we strive to minimise the use of fossil fuels, renewable energy is becoming ever more important. For this reason, wind turbines (which may be on land or offshore) have gained (and continue to gain) interest. Both due to the environment in which they operate and due to their (nonlinear and often difficult to model) dynamics, controlling the operation of offshore wind turbines is challenging. However, for several reasons, it is of interest to ensure that a wind turbine remains at (or close to) specific operating conditions (to minimise structural damage and maximise energy generation). These objectives can be elegantly captured via the formulation of an optimal control problem. Nonlinear optimal control problems are notoriously difficult to solve, however. Therefore, most existing techniques for solving optimal control problems involving (offshore and land-based) wind turbines rely on a linear approximation of wind turbine dynamics, resulting in a so-called Linear Quadratic Regulator (LQR) problem.

Recently, we have developed an alternative technique to solve (nonlinear) optimal control problems approximately. The technique has been demonstrated (via simulations in Matlab) on a nonlinear (albeit simplified) model of an offshore wind turbine with promising results (i.e. improved performance with respect to an LQR approximation). In this project the student will explore how the control technique will perform in more “realistic” settings, e.g. via the use of high-fidelity simulation tools.

In this project the student will consider optimal control of wind turbines. The first objective of the project will be modeling and simulation (of simplified models) of single/multiple wind turbines in MATLAB. The second objective of the project is then to simulate the behaviour of one/more wind turbine/turbines under the designed control action using existing high-fidelity simulation tools, e.g. those that have been developed by The National Renewable Energy

Laboratory (NREL). The performance of the different control strategies will be compared and contrasted using these simulation tools.

The project will involve the consideration of nonlinear dynamical systems and advanced control techniques. For this reason, an excellent background and a keen interest in mathematics and control theory is absolutely essential. The project will involve a programming in Matlab as well, as learning how to use existing simulation tools (such as those developed by NREL). A solid background/aptitude for programming is therefore also required.

Project no:	IC-ACMTM05
Project title:	Techniques to solve dynamic optimisation problems
Supervisor:	Mylvaganam, Thulasi Dr
Category:	Computational;#Theoretical;#Numerical;#Literature
Software:	Matlab
Confidential:	Yes
Offered to:	
	MSc in Advanced Computational Methods

Optimal control concerns the optimisation (i.e. minimisation/maximisation of a cost/reward) subject to a dynamic constraint (i.e. the dynamics of the system). Namely, given a dynamical system, one seeks to design its control input so that the resulting trajectory is optimal with respect to a certain objective. A classical optimal control problem is the so-called Goddard Problem, which concerns the problem of maximising the peak altitude of a rocket. Differential games represent a similar class of problems, involving a dynamical systems and multiple objectives/control inputs. Such problems – which involve optimisation and quantities that evolve in time - are known as dynamic optimisation problems.

With very few exceptions (e.g. the Linear Quadratic Regulator problem), solving dynamic optimisation problems is notoriously difficult. Different techniques to approximate their solution are therefore called for.

Recently two techniques have been proposed, that combine the complementary perspectives provided by the two main “machineries” to solve dynamic optimisation problems (i.e. Dynamic Programming and Pontryagin’s Minimum Principle), to construct (systematically) approximate solutions of (nonlinear) optimal control problems. In this project the student will consider in detail one/both of these techniques. The student may then delve into applications of these techniques into meaningful engineering problems and/or further developments (from a fundamental control theoretic perspective) of these techniques. The performance of designed controllers will be assessed via simulations in Matlab.

This is an ambitious and challenging project that is focused on fundamental control engineering. The project will necessarily involve nonlinear dynamical systems and advanced nonlinear control techniques. For this reason, an excellent background and a keen interest in mathematics and control theory is absolutely essential. A solid grasp of linear control theory is essential and some prior knowledge/exposure to more advanced control theory (optimal control, nonlinear systems, etc.) would be highly beneficial. The student will be required to explore fundamental concepts and topics in nonlinear systems and nonlinear control. The project will involve a programming in Matlab. A solid background/aptitude for programming is therefore also beneficial.

Note that due to the focus of this project on fundamental control engineering, it is well suited for a student with particular interests in this area and general interest in fundamental mathematics, with the relevant background as detailed above.

Project no:	IC-ACMTM06
Project title:	Data-based control with applications to space systems
Supervisor:	Mylvaganam, Thulasi Dr
Category:	Computational;#Theoretical;#Analysis;#Survey;#Numerical;#Literature
Software:	Matlab
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Many “classical” control techniques rely on the availability of a mathematical model which describes how a dynamical system behaves. The availability of accurate models is, however, often limited. In the absence of such a model one may resort to system identification or machine learning techniques. In the context of control these techniques have certain drawbacks; a databased control method has recently been developed which overcomes some of these drawbacks. In this project this data-based method for control design will be considered. In particular, its application to space systems will be explored. The performance of this specific data-based method will be compared and contrasted with other (data- or model-based) methods. The project will involve the consideration of dynamical systems and control techniques (possibly nonlinear), as well as convex optimisation. For this reason, a strong background in and a keen interest in mathematics and control theory is absolutely essential. The project will involve a thorough literature survey related to control of space systems and it will also involve a significant amount of programming in Matlab.

Project no:	IC-ACMTM07/E
Project title:	Data-based control with applications to robotics
Supervisor:	Mylvaganam, Thulasi Dr
Category:	Experimental
Software:	Matlab
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Many “classical” control techniques rely on the availability of a mathematical model which describes how a dynamical system behaves. The availability of accurate models is, however, often limited. In the absence of such a model one may resort to system identification or machine learning techniques. In the context of control these techniques have certain drawbacks; a databased control method has recently been developed which overcomes

some of these drawbacks. In this project this data-based method for control design will be considered. In particular, its application to robotics will be explored. Purely robotic systems and human-robot systems will be considered (the latter is e.g. of interest to robot-assisted therapy). The performance of the specific data-based method will be compared and contrasted with other (data- or model-based) methods. The project will involve the consideration of dynamical systems and control techniques (possibly nonlinear), as well as convex optimisation. For this reason, a strong background in and a keen interest in mathematics and control theory is absolutely essential. The project will involve a thorough literature survey related to control of robotic systems and it will also involve a significant amount of programming in Matlab.

Palacios, Rafael Prof

Project no:	IC-MSCRP01
Project title:	Conceptual design of a hydrogen-powered long-range aircraft
Supervisor:	Palacios, Rafael Prof
Category:	Computational;#Design;#Numerical;#Literature
Software:	python
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Hydrogen may be the solution to achieve net-zero aviation, although there are many technological questions that need addressing. We also need suitable design tools to size vehicles with this new power-system. For long-range flight, direct hydrogen combustion may be available, and the major challenge is regarding hydrogen storage, but in terms of safety and volume constraints. This project will explore this problem, using SUAVE, a state-of-the-art open source conceptual design environment (<https://suave.stanford.edu/>) and OpenVSP, a vehicle parametrization engine (<http://openvsp.org/>).

Project no:	IC-MSCRP02
Project title:	Real-time simulation of flexible vehicles
Supervisor:	Palacios, Rafael Prof
Category:	Computational
Software:	python, C++
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Flight simulation is one of the basic building-blocks in the development of new vehicles, and the training of their crew. Flight simulators however have long been restricted to 6-DOF descriptions of the aircraft dynamics, where the aircraft can be characterised as a rigid-body. However, many concepts for next-generation transport aircraft are considering wings of very long aspect ratio, which display high flexibility during operation. They would need new

strategies real-time flight simulation and those will be investigated in this project. An open-source flight simulator (FlightGear) will provide the interface, and the project will seek to replace its numerical engine (JSBSim) by Imperial's aeroelastic simulator (SHARPy), for which a UDP interface is already available. Acceleration methods based on model reduction will finally serve to bring the simulator to real time operation.

Project no:	IC-MSCR03
Project title:	Conceptual design of a hydrogen-powered regional aircraft
Supervisor:	Palacios, Rafael Prof
Category:	Computational;#Design;#Numerical;#Literature
Software:	python
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Hydrogen may be the solution to achieve net-zero aviation, although there are many technological questions that need addressing. We also need suitable design tools to size vehicles with this new power-system. This project will explore this last point, using SUAVE, a state-of-the-art open source conceptual design environment (<https://suave.stanford.edu/>) and OpenVSP, a vehicle parametrization engine (<http://openvsp.org/>). A new propulsion system will need to be developed, to incorporate hydrogen fuel cells, for exploration of the design space of next-generation zero-emissions regional aircraft.

Project no:	IC-MSCR04
Project title:	Dynamics of flexible vehicles in atmospheric turbulence
Supervisor:	Palacios, Rafael Prof
Category:	Computational;#Numerical
Software:	Python, Linux
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project will address the problem of the numerical prediction of the response to atmospheric turbulence of air vehicles with high-aspect-ratio wings. This is a key problem in the design of ultra-efficient next-generation aircraft, and implies addressing a multi-physics problem on the interaction between a typically very flexible airframe and the non-stationary aerodynamic forces generated through both the atmospheric conditions and the vehicle response to them. In this project, a very light flying wing aeroelastic model will be used to investigate the response to atmospheric turbulence. The classical theory of isotropic turbulence will be used to generate the excitations and the wing response will be studied using methods for dynamical systems analysis. The project will make extensive use of Python. Further reading: <http://dx.doi.org/10.2514/1.G000715>

Project no:	IC-ACMRP05
Project title:	Computational architecture for high-fidelity fluid-structure interaction analysis
Supervisor:	Palacios, Rafael Prof
Category:	Computational;#Numerical
Software:	C++, access to the HPC
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Most aerodynamic surfaces have a non-negligible level of deformation and vibration which affects their performance. This is dealt in design by building computational models for Fluid-Structure Interaction (FSI). In general, that requires separate integration of the time-domain solution of the Navier-Stokes equations on the fluid and the elastodynamic equations of motion on the solid, which are coupled together by means of spatial and time interpolation at the solid/fluid interface. Details on the spatial interpolation are well-known for steady analysis, but this is less clear for vibration analysis, such as wing flutter. This will be the focus of this project, which will investigate strategies for interpolation for implementation in the open-source SU2 code (www.stanford.edu/su2), whose FSI module has been developed at Imperial. This project will make extensive use of the C++. No previous knowledge of the language is required, but excellent programming skills are a must. Further reading: <http://dx.doi.org/10.1002/nme.5700>

Project no:	IC-MSCR06
Project title:	Aerodynamic shape optimization of compliant wings
Supervisor:	Palacios, Rafael Prof
Category:	Computational;#Numerical
Software:	C++, access to the HPC
Confidential:	No
Offered to:	
	MSc in Advanced Computational Methods

Aerodynamic shape optimization using high-fidelity tools is now routinely performed to identify optimum aerofoil shapes and wing planforms. A key pending challenge is the integration of aeroelastic effects on the optimization, since wing deformations fundamentally alter its performance. A potential strategy for this, which is being investigated by researchers at Imperial, is to use algorithmic differentiation strategies to approximate gradients automatically through the manipulation of a computer code without having to derive (and code) any analytical expressions. This has been implemented in the open-source SU2 code (<https://su2code.github.io>), parts of which have been developed at Imperial. The project will investigate the potential advantages of compliant aerofoils for designs with multiple operating points (multiobjective optimization). The project will make extensive use of the C++. No previous knowledge of the language is required, but excellent programming skills are a must. Further reading: <http://dx.doi.org/10.1007/s00158-022-03234-9>

Project no:	IC-MSCR07/E
Project title:	Wind tunnel demonstration of flutter suppression system
Supervisor:	Palacios, Rafael Prof
Category:	Experimental;#Computational
Software:	Matlab, python
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project is an investigation on aeroservoelastic vehicle design, in particular, the integration of control systems on wings to increase their flutter speed (an stability augmentation control problem). This can tilt the trade-off between aerodynamic and structural constraints towards much lighter wing designs, but at the expense of a much complex design evaluation. In this project, flutter suppression strategies will be first numerically investigated on an existing actuated composite wing. A flap driven by an Arduino card has been shown to produce efficient actuation for a wind-tunnel implementation with minimum delays, but the optimal layout of sensors (accelerometers and strain gauges) and the identification of the closed-loop system dynamics needs to be identified. The project will address those issues to identify and demonstrate effective aeroelastic stability augmentation strategies, both through simulation and experiments.

Project no:	IC-MSCR08
Project title:	Understanding uncertainty in dynamic aeroelasticity
Supervisor:	Palacios, Rafael Prof
Category:	Computational;#Numerical;#Literature
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Dynamic aeroelasticity is mainly concerned with the stability, response to atmospheric conditions, and control of flexible aircraft. While deterministic design methods are by now well established, there has been much less effort on quantifying the effect that uncertainty on the model parameters have on the ability on our computational predictions. In this project this will be explored using analytical models for dynamic aeroelastic behaviour and stochastic methods of analysis. This project will be carried out in Matlab.

Further reading: <https://dx.doi.org/10.1146/annurev-fluid-122414-034441>

Panesar, Ajit Dr

Project no:	IC-MSCAP01
Project title:	Sustainability through Additive Manufacturing considering life cycle analysis
Supervisor:	Panesar, Ajit Dr
Category:	Computational;Analysis;Literature
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

There is a growing interest and even regulatory requirements for sustainable manufacturing. While AM has been hailed as the more sustainable option for its additive nature, there has not been much conclusive work in quantifying the impact of various design considerations over the whole product life cycle of AM parts.

This project will aim to develop the quantitative model that assesses the sustainability and economics of additive manufacturing. Potential considerations in the assessment include decentralised manufacturing, CO2 emissions, cost modelling, packing optimisation etc. Successful implementation will allow sustainability considerations to be incorporated into the design optimisation for additive manufacturing process, creating products that are more environmentally friendly and economical.

A strong interest in Machine Learning, Optimisation, and programming in Python is desirable.

Project no:	IC-MSCAP02
Project title:	Intelligent design tool to realise system embodied within a structure
Supervisor:	Panesar, Ajit Dr
Category:	Computational;Design;Analysis;Numerical
Software:	Matlab
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

In collaboration with AMRC Catapult, this project aims to advance the area of multi-material AM. Development of the next generation of products, such as autonomous flying vehicles, require both optimised structures and system elements (<https://www.youtube.com/watch?v=roB5wCr9ITo>) which can be realised using multi-material multifunctional 3D printing to exhibit unrivalled performance.

A design framework has been developed for the creation of a routing system within the structure. This design tool is developed to enable fast and effective manufacturing using the approximation method along the medial axis interpolated by a series of piecewise Catmull-Rom splines.

The contribution of this project is to enhance the design tool to the front-end design tool in Jupyter Notebook, assessing the suitable routing method and generating the CAD model from the design tool ready to be realised. Depending on the student's interested applications, Finite Element analysis or experiment on printed samples can be conducted by considering a) the effect of the reinforcement's thickness b) heat dissipation of electronic components c) evaluating the design performance through conducting experiments on printed reinforced samples.

Programming Language/Software: Python, Abaqus

Project no:	IC-MSCAP03/E
Project title:	Fibre reinforced additive manufacturing
Supervisor:	Panesar, Ajit Dr
Category:	Experimental;Experimental: Structures;Design;Literature
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project offers the student a unique opportunity to work on the state-of-the-art Anisoprint 3D printer to realise high performance fibre reinforced polymeric parts. It is envisaged that besides process parameter optimisation, the student in consultation with the supervisor will be exploring tool-path, infill strategies and fibre volume fractions to enhance the part performance. This experimental work will likely lead to a conference/journal paper.

Project no:	IC-MSCAP04
Project title:	Multiscale modelling for structural power
Supervisor:	Panesar, Ajit Dr (co-supervisor Prof. Emile Greenhalgh)
Category:	Computational
Software:	Python and Abaqus
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Building upon our earlier work on topology optimisation for the multifunctional polymer matrix which are vital for realising structural power composites, we'd like to explore low-fidelity multiscale solutions capable of surpassing performance of current state-of-the-art solutions. Knowledge of Python and Abaqus will prove highly beneficial.

Paranjape, Aditya Dr

Project no:	IC-MSCADP01
Project title:	Control of flutter on moderate-to-high AR wings
Supervisor:	Paranjape, Aditya Dr with Palacios, Rafael Prof
Category:	Mix of theory and computation
Software:	Python; SHARPy
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;#
	MSc in Advanced Computational Methods

Light, moderate-to-high aspect ratio wings are of considerable interest for application to efficient, long-endurance aircraft. These include electric-powered commercial aircraft and long-range, high-altitude remote sensing aircraft. These aircraft are prone to flutter instabilities driven by a coupling between the “rigid body” flight dynamics and the structural dynamics of the wing. The NASA Helios is a case in point.

Our objective is to design a control law for mitigating flutter on moderate-to-high AR wings (AR around 10). We are specifically interested in design techniques that exploit the underlying dynamics described by partial differential equations. Towards that end, we will look to design output feedback control laws and determine their stability margins. While the PDEs can be helpful in deriving analytical stability guarantees, we will also look to use numerical continuation and bifurcation methods to determine the stability margins computationally. We will use a high fidelity model implemented using software called SHARPy developed at Imperial College London.

Note: Dr Paranjape holds an honorary position at Imperial College London. He is a senior scientist at TCS Research, a part of Tata Consultancy Services Ltd, in Pune, India. He will participate remotely.

Prior academic preparation: control theory; partial differential equations; flight mechanics; aeroelasticity.

Peiro, Joaquim Prof

Project no:	IC-MSCJP01
Project title:	Modelling of fluid sloshing using smooth particle hydrodynamics
Supervisor:	Peiro, Joaquim Prof
Category:	Computational
Software:	Matlab or C++ and CUDA (for GPUs).
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering
	MSc in Advanced Computational Methods

The purpose of this work is to assess the suitability of a code, based on the smoothed particle hydrodynamics (SPH) methodology, for modelling fluid sloshing with large amplitude motions; to verify it against other CFD formulations (e.g. volume-of-fluid); and to validate it against available experimental data. Today very few grid dependent or independent CFD techniques are available to accurately simulate this type of flows. SPH is a mesh-free, particle-based formulation for the numerical simulation of physical problems. It is one of the few techniques able to handle complex phenomena such as liquid break-up, fracturing, shattering, and possible phase change etc. It permits an easy implementation of complicated physics such as multiple phases, realistic equations of state, electromagnetic, compressibility, solidification, vaporization, porous media flow and history dependence of material properties. Further, it is able to deal with complex geometries in two and three dimensions.

Project no:	IC-MSJCP02
Project title:	Simulation of air flow and heat transfer in networks of tunnels
Supervisor:	Peiro, Joaquim Prof
Category:	Computational;#Design
Software:	Python, Matlab, and/or C++ or Fortran compilers for projects 1 to 4; StarCCM+ license for project 5.
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering
	MSc in Advanced Computational Methods

Passenger comfort and safety are the main concerns in the design of tunnels. A train travelling at high speed through a tunnel generates a pressure wave that might cause discomfort to the passengers. Similarly, the transit of trains in a network of tunnels generates heat that affects the temperature environment. These are a few of the projects proposed under this theme:

1. Aerodynamic interaction of a train entering a tunnel: A 1D code for the modelling of pressure waves in tunnels will be used to select the optimal tunnel section so that the increase in pressure does not go beyond the threshold of discomfort.
2. Thermal effects during the transit of a high-speed train in a tunnel: To incorporate thermal effects, including fire and thermal conduction in tunnel linings, into an existing aerodynamic code to model thermal effects during normal operation and in the event of a fire.
3. Long term evolution of the temperature in an underground network: To ensure that the fluctuations of temperature within the network are within acceptable limits for passenger comfort. The major challenge is the slow release of heat through the tunnel walls that often requires to model temperature fluctuations over several years.
4. Modeling of tunnel boom cause by micro-pressures waves generated by the entry in the tunnel of trains traveling at high speeds (above 300 km/h). The sound produced at the exit of the tunnel is a function of the pressure history. This will be calculated by the 1D code and the results processed using a suitable model of sound generation from pressure fluctuations.
5. CFD modelling of the pressure wave generated by the entry of a train in a tunnel using StarCCM+. For short tunnels three-dimensional effects are important and the one-dimensional approximation is no longer accurate. This project will compare CFD simulations

with those of the 1D and experimental. These comparisons will be used to validate the CFD simulation and investigate the range of validity of the one-dimensional approach.

Project no:	IC-MSCJP03
Project title:	Simulation of flow in arterial, venous, lymphatic, and respiratory networks
Supervisor:	Peiro, Joaquim Prof
Category:	Computational;#Literature
Software:	Matlab, Python, and C++ or Fortran compilers
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering
	MSc in Advanced Computational Methods

A number of physiological systems, such as the arterial, venous, lymphatic, and respiratory systems, are networks made up of millions of vessels so 3-D flow simulations are just not feasible. Reduced 1-D models of the system based on the area-averaged governing equations provide a computationally affordable and reasonably accurate simulation capability to study the flow in networks of vessels. These are a few of the projects proposed under this theme:

1. A 1-D model of the venous system: To adapt an existing 1-D solver for the arterial system to the venous system by incorporating an appropriate pressure-area relation for veins and a model of valves. Application to cerebral arteriovenous malformation
2. A 1-D model of the respiratory network: To investigate suitable models of the morphology of the bronchial tree and to develop an hydraulic model of the respiratory network. Application to assessment of asthma.
3. Modelling auto-regulation in the arterial system: Use the 1-D model to investigate the mechanisms of regulation of blood flow in the arterial system when it is subject to external actions such as, for instance, changes in temperature, pressure or posture.
4. Modelling of the flow in the lymphatic system: To devise a formulation to model the peristaltic motion of lymph via the contraction of the lymphagions and their valves.

Project no:	IC-MSCJP04
Project title:	Uncertainty quantification for aerodynamic simulation
Supervisor:	Peiro, Joaquim Prof
Category:	Computational
Software:	Python or Matlab, xfoil, possibly CFD codes (e.g. Starccm+)
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering
	MSc in Advanced Computational Methods

To assess the effect of operational and geometrical uncertainty on aerodynamic performance we will use non-intrusive uncertainty quantification methods to compute

probabilistic expectations of quantities of interest, such as lift and drag coefficients, from samples taken simultaneously and independently on a hierarchy of computational meshes. This way most of the computational effort is transferred from the finest mesh to the coarser meshes thus significantly reducing computational cost. We will apply the method to 1D nozzle flows and 2D flows past aerofoils to assess the effect of uncertainty in operational and geometric parameters on performance.

Project no:	IC-MSJJP05
Project title:	Reconstruction of high-order surface data from triangulations
Supervisor:	Peiro, Joaquim Prof
Category:	Computational;#Numerical;#Literature
Software:	Python, C++, Julia, Matlab, or compiler of choice
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering
	MSc in Advanced Computational Methods

The definition of a computational domain for fluid simulation requires an analytical definition of its boundary which is usually generated using a CAD system. However, if the geometry of the domain is obtained via a laser scan or medical imaging, the geometry of the boundary is often given by a triangulation of its surfaces. CFD simulations require the generation of meshes of varying resolution for a given problem for instance to adapt the mesh to a solution or simply to perform mesh convergence studies. In such cases we require a highly accurate, continuous geometric representation of the surface to generate these meshes. This project will review the state-of-the-art of methods for high-order surface reconstruction and implement, in the programming language of your choice, suitable candidate methods that satisfy these requirements: a) the reconstructed surface has a high degree of continuity, i.e. above first order; b) the method is numerically stable and c) it preserves sharp features of the surface and represents the geometry accurately.

Pinho, Silvestre Prof

Project no:	IC-AAESP01
Project title:	A holistic approach to achieve NetZero in aviation by 2050
Supervisor:	Pinho, Silvestre Prof
Category:	Theoretical;#Survey;#Literature
Software:	N/A
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering

Airbus and ZeroAvia expect to have a large demonstrator aircraft fuelled by hydrogen flying by 2035. Even if they are successful, relying solely on these hydrogen-fuelled aircraft to achieve NetZero by 2050 would give us only 15 years to replace all kerosene-fuelled aircraft in the world. This seems unlikely. Achieving NetZero in aviation by 2050 will instead require many factors, including hydrogen propulsion, SAF, e-fuels, carbon capture, demand management, airport infrastructure, green energy generation, distribution, and storage, coordinated policies around the world, etc.

Some of these challenges are purely engineering challenges, but others are psychological, economic and geo-political challenges. Some of them are indirect or not immediately obvious: e.g. reducing demand via green fees could make air travel a privilege of only the rich; any demand management will likely reduce the economic value of the aerospace sector (which could additionally disincentivise research investments for greener aircraft technologies); reducing our dependence on kerosene can destabilise oil-exporting countries; SAF can replace other crops and contribute to famine; Silicon, needed for photovoltaic cells, is mostly produced in one single country; 50% of the world's copper, of which each wind turbine needs several tons, come from only two countries.

In this project, you will review all the factors that can play a role in achieving NetZero in aviation and analyse their direct and indirect effects. Based on this analysis, you will propose several scenarios that use sub-sets of all these factors to achieve NetZero in aviation by 2050, and discuss the relative implications of each scenario in terms of environment, engineering, economics and geo-politics.

Project no:	IC-MSCSP02
Project title:	Overall structural design of a hydrogen-powered aircraft dry-wing
Supervisor:	Pinho, Silvestre Prof
Category:	Computational;#Theoretical;#Design;#Analysis;#Numerical;#Literature
Software:	Abaqus
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

Hydrogen is the most promising technology to allow aviation to meet NetZero by 2050. Airbus carried out an extensive programme (ZEROe) where they proposed three aircraft concepts that could meet NetZero requirement for different market segments. Two of these solutions include dry-wing solutions. In the UK, the Aerospace Technology Institute recently concluded another project (FlyZero) with similar aims where they proposed three new concepts, all of which including dry-wings. In dry-wing solutions, the fuel is not carried inside the wings, and these can therefore be thinner. This poses significant challenges to their internal structure as they need to resist large bending moments with a smaller second moment of area.

In this project, you will start by reviewing the outcome of the FlyZero project, and then proceed to create a CAD and subsequently a FEM model of a dry wing for one of the FlyZero concepts, including its internal structure of stiffeners and ribs and you will design it to resist the applied forces while keeping deformations within an acceptable range.

Project no:	IC-MSCSP03
Project title:	Structural design of a liquid hydrogen tank for an aircraft
Supervisor:	Pinho, Silvestre Prof
Category:	Computational;#Theoretical;#Design;#Analysis;#Numerical;#Literature
Software:	Abaqus
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

Hydrogen is the most promising technology to allow aviation to meet NetZero by 2050. Airbus (ZEROe) and The ATI (FlyZero) carried out extensive programmes (ZEROe and FlyZero respectively) where they proposed in total six aircraft concepts that could meet NetZero requirement for different market segments: all of them are hydrogen-powered, in some cases with liquid hydrogen.

Safe storage of liquid hydrogen in tanks is challenging due to the pressures required, to the low temperatures, and the ease with which hydrogen can leak from any cracks. In this project, you will review various solutions for liquid hydrogen tanks, design your own, create a CAD model and an FE model for it, and finally iterate your solution until you have a solution with suitable structural integrity.

Project no:	IC-MSCSP04/E
Project title:	Design of bio-inspired structures manufactured by Automated Fibre Placement
Supervisor:	Pinho, Silvestre Prof
Category:	Experimental;#Experimental: Structures;#Design
Software:	Abaqus
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

We recently acquired an AFP machine suitable for manufacturing continuous fibre reinforced composites which provides us with unique manufacturing capabilities extremely relevant for industry.

AFP can be used to design and automatically manufacture complex CFRP parts that would otherwise be overly labour intensive or even impossible.

In this project, you will design, simulate, manufacture (using AFP) and test a bio-inspired structure, such as a skin-stiffener structure.

Project no:	IC-MSCSP05
Project title:	Physics-informed AI and Bayesian optimisation for structural analysis
Supervisor:	Pinho, Silvestre Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Abaqus, Tensorflow
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Currently, material models for advanced composite materials used in aircraft structures are based on sophisticated analytical idealisations of the material at very small scales. This physical basis gives engineers some confidence that their structural designs will be sound. However, then manufactured at large scales, materials are often different to those manufactured at small scales. There is therefore a need for developing material models from information arising at larger scales.

Digital Image Correlation allows us to obtain the strain field for a large component being tested, including during failure, all while measuring the external work done by the applied forced.

In this project, you will design, train and test a physics-informed AI model to extract the constitutive response (including during damage growth) from the strain field and external work in one or more simple components. Rather than use experimental strain data (which would be the ultimate aim but is more time consuming), you will create FE models of these simple components to generate the training strain data. You will then train your model and iterate the model architecture until your model is able to predict the constitutive response. Once your ML model is built and validated, you will use Bayesian Optimisation to find out the configuration of the FE model (shape, layup, features such as holes, etc) that maximise the predictive capability of the ML model.

The outcome of this project will be a proposed architecture for the ML model, as well as the definition of the corresponding experimental test specimens that should be manufactured and tested to apply your model in a real scenario.

Project no:	IC-MSCSP06
Project title:	Doubling the service life of a wind turbine blade
Supervisor:	Pinho, Silvestre Prof
Category:	Computational;#Theoretical;#Design;#Analysis;#Numerical;#Literature
Software:	Abaqus
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

Achieving NetZero by 2050 will require expanding very significantly our capability to produce green energy. In this context, we will likely experience exponential growth of wind-turbine farms over the next decades. However, this comes with a problem: despite extensive research, the prospects for recycling or even re-purposing wind-turbine blades are grim. Used wind turbine blades are therefore a large and rapidly growing problem. The expert opinion is that the most effective way to reduce the environmental impact of a turbine blade

is to increase its service life: it is estimated that its service life can be doubled with an increase in mass of about 10%.

In this project, you will analyse the most modern designs for wind turbine blades and how they fail, and propose how to modify the design so that their service life can be increased by a factor of 2.

Project no:	IC-MS CSP07
Project title:	Development of an automated validation infrastructure for advanced failure models
Supervisor:	Pinho, Silvestre Prof
Category:	Computational;#Analysis;#Numerical
Software:	Abaqus
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

Failure models for composites have evolved tremendously in the past decades and continue to evolve. A typical characteristic of current validation procedures is that they tend to be restricted to a few test cases and with models built from scratch every single time. This differs from code validation methodologies in other areas, where an automated and comprehensive code validation and verification is carried out ahead of new versions of a code being committed.

In this project, you will: compile and rationalise a series of test cases; use, modify or build a corresponding FE model (as appropriate); and build a Python code that will take a more recent version of a failure model, run the various test cases (including generating failure envelopes), and create suitable infographics of the most significant results, with an indication of how they differ from reference data.

Project no:	IC-MS CSP08
Project title:	Your own project idea
Supervisor:	Pinho, Silvestre Prof
Category:	Computational;#Theoretical;#Design;#Analysis;#Survey;#Numerical;#Literature
Software:	
Confidential	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

You may have your own idea for a masters project that you would really like to do, for instance either because it addresses a problem you feel strongly about, because it falls under an area that you are passionate about, or because of some experience/knowledge/skills you have that you would like to leverage during your project.

If the above falls broadly under my area of experience (mechanics of composite materials and structures), you can choose this project. My goal will be to enable you to carry out this project of yours.

If this project gets allocated to you, we will meet at the start of the project and discuss its context, goals, originality and methodology. This is so that we define together a detailed plan that can fulfill the requirements of a masters project. Note that, at this stage, we may eventually need to adapt somewhat the details of what you had in mind so that the plan is suitable for a masters. (As a note, in the unlikely scenario that we conclude, during this initial meeting, that your idea cannot possibly be suitable for a masters project, I can offer you a range of possible alternative projects so that, as long as you are broadly interested in structures, you are guaranteed that you will have a suitable project. But I fully expect that not to be needed.)

From that moment, we will update the project title accordingly and this project will continue as any other.

If you are unsure whether an idea you have would be suitable for a masters project and would like to discuss ahead of applying for this project, feel free to email me or to knock on my office door. Note that this will have no effect on whether this project gets allocated to you.

Given the topic of this project is not defined during the project allocation stage, it is not possible to plan experimental resources around this project, and hence the project needs to be purely numerical / theoretical.

Project no:	IC-MSCSP09
Project title:	Development of an automated validation infrastructure for advanced failure models
Supervisor:	Pinho, Silvestre Prof
Category:	Computational;#Analysis;#Numerical
Software:	N/A
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

Failure models for composites have evolved tremendously in the past decades and continue to evolve. A typical characteristic of current validation procedures is that they tend to be restricted to a few test cases and with models built from scratch every single time. This differs from code validation methodologies in other areas, where an automated and comprehensive code validation and verification is carried out ahead of new versions of a code being committed.

In this project, you will: compile and rationalise a series of test cases; use, modify or build a corresponding FE model (as appropriate); and build a Python code that will take a more recent version of a failure model, run the various test cases (including generating failure envelopes), and create suitable infographics of the most significant results, with an indication of how they differ from reference data.

Ribera Vicent, Maria Dr

Project no:	IC-MSCMRV01
Project title:	Development of a comprehensive helicopter flight dynamics code
Supervisor:	Dr Maria Ribera Vicent
Category:	Computational;#Analysis
Software:	FORTRAN
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

A comprehensive helicopter flight dynamics code, capable of modelling both steady-state and transient manoeuvres for a range of flight conditions, is available as a baseline for this project. This code has been expanded on by different contributors to add more features or extend its range of usability. The proposed projects will extend this code by adding new functionality. The precise projects will be defined based on the evolution of the code at the point of starting the project, but possible examples include: looking at the effect of rotor in-plane induced velocities on manoeuvres, analysing the interference of the main rotor wake on the tail rotor, modelling the rotor wake aerodynamics in ground effect, improving the aerodynamic modelling of the fuselage and empennage, etc. This is a large legacy code written in FORTRAN. Excellent programming skills are essential, FORTRAN programming skill are desirable but can be learned at the start of the project by a competent programmer.

Project no:	IC-MSCMRV02
Project title:	A panel method for rotor/fuselage interaction in a helicopter flight dynamics simulation model
Supervisor:	Dr Maria Ribera Vicent
Category:	Computational;#Analysis
Software:	FORTRAN (or very good programming skills with ability to learn FORTRAN quickly) is an absolute requirement. CAD/Meshing software of choice.
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

A free-vortex rotor wake model can improve the aerodynamic modelling of a helicopter and improve the predictions of the dynamic response using a comprehensive flight dynamics model. While CFD, and even more CFD/CSD, can very accurately capture the interaction between rotor and fuselage, it comes at a large computational cost, which might not be realistic for flight dynamics applications. The proposed project will couple a panel method model of the UH-60 helicopter with a flight dynamics model which includes refined rotor wake aerodynamics (time-accurate free-wake model) and evaluate whether the new fuselage modelling improves the trim validation with respect to flight test data.

Project no:	IC-MSCMRV03
Project title:	Effect of Main Rotor – Tail Rotor wake interaction on the yaw performance of a UH-60 helicopter
Supervisor:	Dr Maria Ribera Vicent
Category:	Computational;#Analysis
Software:	FORTTRAN (or advanced programming skills that will let you learn program quickly) is required, MATLAB
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

The aerodynamic environment of a helicopter is very complex, with the main rotor wake interacting with the fuselage, empennage and tail rotor wake. A comprehensive helicopter flight dynamics model is available which has been validated against the UH-60 flight test program. This code includes a time-marching rotor vortex wake model which is capable of capturing the wake dynamics during manoeuvres and allows the wake vortices to distort under their mutual influence. The tail rotor, however, is modelled using a simpler dynamic inflow model, and the interaction between main rotor and tail rotor wakes is not captured. While the flight dynamics model correlates well with the flight test data in general, the tail rotor control is poorly predicted. This project aims to explore the interaction between the main and tail rotor wakes and provide insights into whether the yaw performance of the model can be improved by allowing for the effect of the main rotor on the tail rotor induced velocities.

Rigas, Georgios Dr

Project no:	IC-MSCGR01/E
Project title:	Experimental reinforcement learning on-chip
Supervisor:	Dr Georgios Rigas
Category:	Experimental;#Numerical
Software:	Labview, TensorFlow, TensorForce
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Reinforcement Learning (RL) is an area of machine learning concerned with how software agents ought to take actions in an environment in order to maximize the notion of cumulative reward. During the last years, we have been using this technique for discovering optimal flow control strategies for aerodynamic drag reduction. However, this framework has been demonstrated using numerical simulations. For implementation on a realistic configuration (i.e car with controllable aerodynamic surfaces), a hardware implementation of the above RL algorithms is required and their execution on Real-Time. In this project, we will use state-of-the-art hardware for real time processing from National Instruments to interface and implement RL agents that are calculated on TensorFlow/TensorForce.

Project no:	IC-MSCGR02
Project title:	Reinforcement learning for flow control
Supervisor:	Dr Georgios Rigas
Category:	Computational;#Analysis
Software:	Python, TensorFlow, TensorForce
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

In this study, we will exploit modern reinforcement learning algorithms for discovering optimal control laws in separated flows, relevant to the automotive and aerospace industry. The demonstration will be performed using simplified 2D Direct Numerical Simulations. The framework/code for RL and the DNS for bluff body flows already exists.

Prerequisites: experience with Python (and preferably Tensorflow).

Project no:	IC-MSCGR03
Project title:	Symmetry break of 3D road vehicle wakes while cornering
Supervisor:	Dr Georgios Rigas
Category:	Computational;#Numerical
Software:	C++, FreeFem++
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Hydrodynamic instabilities in the wake of bluff-bodies (such as vortex shedding) are responsible for aerodynamic drag, structural fatigue and noise. We have calculated these instabilities computationally in 3D geometries with 3 inhomogeneous directions using massively parallel eigenvalue computations and HPC.

In this study, we will calculate and investigate the 3D bifurcations and instabilities in the wake of a 3D bluff body for different angles of attack (yaw angles). Preliminary results have shown that an unstable degenerate eignmode splits into two distinct modes with incommensurate frequencies when the yaw angle is increasing, which could explain results observed in a variety of 3D fluid flows when the underlying symmetry is broken.

Project no:	IC-MSCGR04
Project title:	Data-driven hydrodynamic stability analysis
Supervisor:	Dr Georgios Rigas
Category:	Computational;#Theoretical
Software:	TensorFlow
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

During the last decade, we have extensively used Proper Orthogonal Decomposition (POD) and variants of this decomposition method (known also as Principal Component Analysis-PCA) for feature extraction. This method has enabled the discovery of coherent structures that dictate the dynamics of flow behind road vehicles, jet engines and wall bounded flows. However, typically a large number of POD/PCA modes is required to capture the dynamics of the inherently nonlinear flow. Nowadays, machine-learning algorithms offer a new paradigm for investigating complex systems. The POD, or linear PCA, can be formulated as a two-layer Neural Network (an autoencoder) with a linear activation function for its linearly weighted input, which can be trained by stochastic gradient. This formulation is an algorithmic alternative to linear eigenvalue/eigenvector problems in terms of NNs, and it offers a direct route to the nonlinear regime and deep learning by adding more layers and a nonlinear activation function on the network.

In this project we will explore this novel framework in order to generate an optimal basis of modes to describe the flow dynamics.

Project no:	IC-MSCGR05
Project title:	Aerodynamics for road-vehicle platooning
Supervisor:	Dr Georgios Rigas
Category:	Computational;#Analysis
Software:	In house DNS code.
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Car platooning is the linking of two or more cars in convoy, using automated driving support systems.

In this study, we will investigate the aerodynamics of 2 or more vehicles during platooning. The demonstration will be performed using simplified 2D Direct Numerical Simulations. The relative distance between initially two cars will be varied and their aerodynamic performance and instabilities will be investigated. The framework for the simulation (DNS) for one bluff body already exists. Prerequisite: use of terminal and Unix.

Prerequisite: use of terminal and Unix.

Project no:	IC-MSGR06
Project title:	Flow instabilities in the wake of realistic car geometries
Supervisor:	Dr Georgios Rigas
Category:	Computational;#Numerical
Software:	C++, FreeFem
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Hydrodynamic instabilities in the wake of bluff-bodies (such as vortex shedding) are responsible for aerodynamic drag, structural fatigue and noise. We have demonstrated these instabilities computationally in simplified geometries with one or more homogeneous directions. In this study, we will apply recently developed numerical tools to fully three-dimensional geometries of industrial relevance to examine their hydrodynamic stability. Specifically, we will investigate the 3D bifurcations and instabilities occurring in the wake of the Ahmed body, a widely accepted benchmark geometry used in the automotive industry. Preferably basic knowledge of C or C++, Unix environment. The code for the 3D stability analysis is already available and validated for canonical 3D geometries.

Preferably basic knowledge of C or C++, Unix environment. The code for the 3D stability analysis is already available and validated for canonical 3D geometries.

Robinson, Paul Prof.

Project no:	IC-MSCPR01/E
Project title:	Investigating the behaviour of shape memory composites for deployable structure applications
Supervisor:	Robinson, Paul Prof
Category:	Experimental, Structures
Software:	N/A
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

An interleaved composites has been developed in the College which can exhibit controllable flexural stiffness and shape memory capability. This material may find uses in deployable, collapsible and morphing / adaptive shape structures. This project will devise, manufacture and investigate simple deployable configurations and will investigate whether it is possible to 'retrain' a configuration to adopt a different memorised shape.

Project no:	IC-MSCPR02/E
Project title:	Development and Investigation of a novel Mode I interlaminar toughness test with improved stability
Supervisor:	Robinson, Paul Prof
Category:	Experimental;#Computational
Software:	ABAQUS
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

Interlaminar toughness testing of composites laminates is most commonly performed using the double cantilever beam method. This method works well for laminates which show no strong variations in toughness as the delamination front advances along the specimen. However some materials, such as woven reinforced composites, exhibit variations in toughness which can result in unstable crack growth in the standard DCB. For such materials it is difficult to measure the toughness variation correctly – often only the high toughness values are determined. A modified loading jig has been proposed to improve the crack growth stability and a previous project has performed some initial investigation. This modified loading jig will be further developed and investigated in this project. The investigations will include finite element modelling and experimental work.

Project no:	IC-MSCPR03/E
Project title:	Investigation of an interleaved strategy for easy repair of impact damage in composites
Supervisor:	Robinson, Paul Prof
Category:	Experimental
Software:	ABAQUS
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

Impact damage in a laminated composite often consists of interlaminar fracture between the laminae and repair of this damage usually involves removal of the damaged material by machining and adhesive bonding of a repair patch. Recent work at Imperial has explored an easy repair concept in which a thermoplastic interleaf is added to the composite to provide a repairable 'weak link'. On impact the properties of the interleaf are chosen such that the damage occurs preferentially within the interleaf. The composite can then be heated and the damage repaired. This project will perform further experimental investigations to help develop and assess this easy repair concept.

Project no:	IC-MSCPR04/E
Project title:	Experimental investigation of strategies for improving the compression performance of carbon fibre composites
Supervisor:	Robinson, Paul Prof
Category:	Experimental;#Experimental: Structures;
Software:	ABAQUS
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

Unidirectional composites subjected to compression in the fibre direction will commonly fail by the development and growth of a kink band. In this project experimental techniques will be used to observe the kink band initiation and growth and to develop & investigate strategies to delay this process.

Project no:	IC-MSCPR05/E
Project title:	Experimental investigation of strategies for improving the resistance of laminated composites to delamination
Supervisor:	Robinson, Paul Prof
Category:	Experimental;#Experimental: Structures; Computational
Software:	ABAQUS
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

Laminated polymer matrix composites are susceptible to delamination (i.e. crack growth between the laminae). In this project strategies to improve the resistance to delamination will be investigated using finite element modelling and experiments. For example, some composites possess a high resistance to delamination because they develop a considerable amount of fibres which bridge between the delaminated surfaces in the wake of the crack tip. Experiments will be conducted to investigate methods for modifying the composite to ensure that extensive fibre bridging or multiple delaminations will occur in the event of the growth of a single delamination.

Santer, Matthew Dr

Project no:	IC-AAEMS01/E
Project title:	Characterization of Dynamic Roughness from Membrane Substrate Buckling
Supervisor:	Dr Matthew Santer
Category:	Experimental: Ready-made experiment;#Computational;#Analysis
Software:	Abaqus
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

When a stiff membrane attached to a relatively-compliant substrate is subjected to compressive stress, interesting patterns appear in the surface due to constrained buckling of the membrane. These patterns are consistent, recoverable and have many potential forms depending on the applied stress. Such a structure has significant potential for use as a flow control device by allowing an aerodynamic surface to transition from smooth to rough and back. Building on existing work this project will characterize the patterns via a combination of finite element analysis using Abaqus and by experimental measurement using image-based techniques. In particular the project will investigate the impact of edge effects and dynamics on the achieved patterns in comparison to the theoretical predictions.

Project no:	IC-AAEMS02
Project title:	Dynamic Analysis for Launch of the HATHOR Deployable Aeroshell
Supervisor:	Dr Matthew Santer
Category:	Computational;#Analysis
Software:	Abaqus
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

The Hypersonic Foldable Aeroshell For Thermal Protection Using Origami (HATHOR) has been designed and developed at Imperial College and is the subject of on-going design and test work. One crucial aspect of the design that has not yet been considered, and which will form the basis of this project, is the characterization of the response of the structure to the large loads imparted during launch. Using the Abaqus finite element analysis package the aeroshell will be modelled in its stowed configuration and the response to launch loads (quasi-static, sinusoidal and random) will be assessed. On the basis of this analysis, modifications to the design will be proposed to ensure increased robustness to the launch environment and key accelerometer placements will be determined to enable validation in a subsequent shaker experiment.

Project no:	IC-AAEMS03
Project title:	Deployable Structures using Bistable Tetrahedra
Supervisor:	Dr Matthew Santer
Category:	Computational;#Design
Software:	Abaqus
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

Previous research has demonstrated how combining nonlinear tape springs with linear leaf springs can enable deployable bistable tetrahedral structures with a substantial change in volume between the stable states. Using numerical modelling using the Abaqus finite element package this project will determine viable actuation methodologies to allow the bistable structure to transition repeatedly between its stable configurations. In addition

methodologies for connecting the bistable structures together will be investigated. Using the principles of digital robotics the resulting structures will be highly reconfigurable with application for spacecraft antennas.

Project no:	IC-AAEMS04/E
Project title:	Deployable Structures for Small Satellites
Supervisor:	Dr Matthew Santer
Category:	Experimental;#Computational
Software:	Abaqus
Confidential:	Yes
Offered to:	MSc in Advanced Aeronautical Engineering

Two open-ended general projects are offered on the design and demonstration of small-satellite scale deployable structures. The first stage will be to produce a specification based on a clear commercial or scientific goal — optics for Earth Observation, Synthetic Aperture Radar, Solar Sails, etc. Design will be carried out of an appropriate deployable structure based on heritage concepts but effectively miniaturized to the required scale. The structure will be required to deploy in a 1g environment for ground testing purposes. This will require a combination of numerical (finite element) modelling using the Abaqus package, and analytical techniques for validation and preliminary assessment. A benchtop functional model will be fabricated and used for verification and validation.

Project no:	IC-AAEMS05
Project title:	Optimal multifunctional thermal protection systems for hypersonic vehicles
Supervisor:	Dr Matthew Santer
Category:	Numerical;#Computational
Software:	Fenics (python)
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering
	MSc in Advanced Computational Methods

Next generation hypersonic vehicles are subjected to extreme heat fluxes over several minutes. It is of great interest to design combined structure and thermal protection systems in order to achieve lightweight architectures which can sustain the combination of high temperatures and thermal stresses and protect internal components during flight.

This project will carry out adjoint-based time-dependent optimization of candidate material combinations subjected to flight loads and heat fluxes. The optimization and thermo-structural finite element analysis will be carried out using the Python-based PDE solver Fenics. The aim of the project is to identify promising architectures and material combinations, and to reveal the underlying physics of the interactions between thermal and structural performance for this challenging and timely problem.

Shamsuddin, Siti Rosminah, Dr

Project no:	IC-COMSRS01/E
Project title:	Understanding thermal-mechanical behaviour of UD and quasi-isotropic carbon fibre reinforced epoxy composites with thermoplastic interleaving
Supervisor:	Shamsuddin, Siti Rosminah Dr
Category:	Experimental;#Theoretical;#Design;#Analysis;#Numerical
Software:	
Confidential:	No
Offered to:	
	MSc in Composites

Carbon fibre reinforced polymer composites have been extensively used as high-performance material for aviation and aerospace application. However, the mechanical property of this composite is limited at elevated temperature because it is governed by the polymer used. The objective of this study is to manufacture various composite designs and analyse the impact of fibre orientation, interleaving ratio, interleave polymers on the composite's thermo-mechanical properties. The experimental data will be used to predict composite's creep and linear viscoelastic properties. This project is open to a maximum of 2 students.

Project no:	IC-COMSRS02/E
Project title:	Impact of mechanical properties on UD carbon fibre reinforced epoxy composites with thermoplastic interleaving
Supervisor:	Shamsuddin, Siti Rosminah Dr
Category:	Experimental;#Design;#Analysis
Software:	
Confidential:	No
Offered to:	
	MSc in Composites

Composites offer advanced design possibilities, improved safety and extended service life with respect to its resistance to corrosion as well as reduced through-life cost. Thermoset-based polymer composites have dominated the market since 1960, and many attempts have been made to improve the matrix's resistance to initiate and propagate cracks. In this study, UD carbon fibre composites will be interleaved with a thermoplastic film. The effect of fibre orientation, interleaving design, thickness of the thermoplastic film, and varying the thermoplastic material on the composite's mechanical properties will be analysed. This include but not limited to: tensile, compression, shear, bending and impact properties. This project is open to a maximum of 3 students.

Project no:	IC-COMSR03/E
Project title:	Interlaminar fracture behaviour of thermoplastic interleaving thermoset composites
Supervisor:	Shamsuddin, Siti Rosminah Dr
Category:	Experimental;#Theoretical;#Design;#Analysis;#Literature
Software:	
Confidential:	No
Offered to:	
	MSc in Composites

Advanced fibre reinforced polymer composite materials are often used in load-bearing and structural applications due to their high strength, high modulus and low density. Interleaved composites have been studied extensively as composite toughening or composites with controllable stiffness. This study is aimed to explore the interlaminar fracture toughness of composites. Two variations of thermoplastic interleaved carbon fibre reinforced polymer (Epoxy) (CFRP) composites will be analysed: where the interface is between interleaved and CFRP and where the interface is between two thermoplastic interleaved films. The composites will also be subjected to 4-point bend and short beam shear tests to study their fracture behaviour. A different approach would be to study different interleaving polymer films with various thicknesses. This project is open to a maximum of 2 students.

Sharif Khodaei, Zahra Prof

Project no:	IC-AAEZSK01/E
Project title:	Development of a visualisation platform for structural health monitoring application
Supervisor:	Sharif Khodaei, Zahra Prof
Category:	Experimental: Ready-made experiment;#Computational;#Analysis
Software:	Matlab or Python
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering

Structural health monitoring (SHM) platform consists of data acquisition, data processing and visualisation to result in structural diagnosis and consequently prognosis. SHM systems consist of sensorising engineering structures and through gathering and processing of the sensor data remotely, assess the health and integrity of the structure. There are many methodologies available for data processing which results in damage detection and characterisation. However, for the maintenance engineer to be able to take action, the location and severity of damage has to be visualised. Therefore, there is a need to develop a visualisation platform which takes the SHM sensor data as an input and outputs the map of the structure with the possible damage location. This project is focused on developing an imaging algorithm which visualises the location and intensity of damage by different post-processing of the collected sensor data. There are experimental sensor data available from previous projects, and the student will use these data to develop an algorithm for detecting and visualising damage. An excellent knowledge of programming such as Matlab or Python is essential, and it is highly recommended to take the structural integrity and health monitoring module as a pre-requisite to this project.

Project no:	IC-MSCZSK02
Project title:	Metamodel development for Active or Passive sensing methods in metallic or composite plates – Project 1
Supervisor:	Sharif Khodaei, Zahra Prof
Category:	Computational;#Theoretical;#Analysis;#Numerical
Software:	Matlab, Python and Abaqus
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Structural health monitoring technique involves detecting damage in a sensorized structure based on the sensor recorded data. There are various ways that the sensor data can be used in structural diagnosis and prognosis, such as developing a meta-model. The purpose of this project is to investigate various meta-modelling techniques for large amount of data that is acquired from sensors and compare their prediction capabilities with each other. There is already a large database of impact data (recorded sensors signals) as well as guided wave data (Active sensing) from various impact and damage scenarios on composite plates. These experimental data will be used for the development of meta models. Some examples of such meta-modelling techniques are response surfaces, or machine learning. This project can be offered to 2 students, one focused on active and one focused on passive sensing applications. This project requires excellent knowledge of Matlab/Python and Abaqus and basic knowledge of signal processing, machine learning and structural dynamics.

Attending the Structural Integrity and Health Monitoring module would be highly recommended for this project.

Project no:	IC-COMZSK03
Project title:	Metamodel development for Active or Passive sensing methods in metallic or composite plates – Project 2
Supervisor:	Sharif Khodaei, Zahra Prof
Category:	Computational;#Theoretical;#Analysis;#Numerical
Software:	Matlab, Python and Abaqus
Confidential:	No
Offered to:	
	MSc in Composites

See project description above.

Project no:	IC-COMZSK04
Project title:	Critical assessment of methodologies and technologies for early detection of material degradation in composite structures
Supervisor:	Sharif Khodaei, Zahra Prof
Category:	Theoretical;#Analysis;#Survey;#Numerical;#Literature
Software:	Matlab and Abaqus
Confidential:	No
Offered to:	
	MSc in Composites

In recent years more and more engineering structures are being made of composites, in particular due to weight saving which can result in greener structures. One particular application is in pressure vessels such as fuel tanks for aerospace applications. One big challenge for the application of various types of composite for pressure vessels is the high load combined with high temperature which can result in material degradation. Early detection of degradation in composite with existing non-destructive inspection techniques is very difficult, if not impossible. Therefore structural health monitoring techniques might offer a solution here. This project is focused on critically investigating both possible sensor technologies (such as fibre optic sensors, strain sensors, capacitance sensors) as well as methodologies (non-linear acoustics) for detecting material degradation at early stages. It will require a thorough literature review and critically assessing the existing solution, combined with some modelling and theoretical assessment. Some numerical modelling will also be carried out with Abaqus, and some data processing is required with programming tools such as Matlab or Python.

Project no:	IC-COMZSK05/E
Project title:	Damage detection in composite structures based on structural health monitoring
Supervisor:	Sharif Khodaei, Zahra Prof
Category:	Experimental;#Computational;#Analysis
Software:	Abaqus, Matlab/Python (ESSENTIAL), Labview
Confidential:	No
Offered to:	
	MSc in Composites

The purpose of the project is to detect in Barely Visible Impact Damage (BVID) in composite structures using Structural Health Monitoring techniques. The SHM technique is based on an array of Piezoelectric (PZT) transducers, actuating Lamb waves in the structure and sensing the waves after propagating. The changes in the propagation properties due to damage can be used for detecting and characterising the damage. There are various algorithms developed for detecting and characterising damage based on comparing the propagational properties of the wave in pristine and damage state. However, their applicability to complex geometries remains questionable. The experiment will include sensorizing a composite structure. Afterwards the structure is impacted to cause BVID. The PZT transducers are used to interrogate the structure before and after each impact. The theoretical part of the project will be modelling wave propagation numerically using Abaqus and to validate the actuation and sensing signals against the experiment. The sensor signals are then used for

development and assessment of appropriate damage detection algorithms based on Lamb waves to detect, locate and characterize damage. This project will require excellent knowledge of programming in Matlab/Python (essential) and good knowledge of Abaqus. Attending the structural integrity and health monitoring module is a pre-requisite to this project.

Project no:	IC-MSZSK06
Project title:	Probability of Detection (PoD) analysis for damage detection in metallic structures
Supervisor:	Sharif Khodaei, Zahra Prof
Category:	Computational;#Theoretical;#Numerical
Software:	Essential: Matlab, R, Python, Abaqus
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

The successful deployment of systems for health monitoring of structures depends on appropriate verification and validation (V&V) of these structural health monitoring (SHM) systems. The V&V method must explicitly evaluate all aspects of the SHM system that can affect its capability to detect, localize, or characterize damage. Moreover, it must evaluate the effects that usage and environmental conditions have on these capabilities over time. A critical part of the assessment is definition and determination of probability of detection (Pod) based on either experimental or numerical data. This project is aimed at investigation and developing PoD methods for damage detection in metallic structures using statistical and reliability based methods combined with numerical modelling. It requires excellent knowledge of non-linear modelling in Abaqus, Fracture mechanics for fatigue crack growth in metallic plates and programming in Matlab together with Python. Attending the structural integrity and health monitoring module would be highly beneficial for this project.

Project no:	IC-COMZSK07
Project title:	Critical Assessment of Damage Detection Methodologies for Structural Health Monitoring in Composite Structures
Supervisor:	Sharif Khodaei, Zahra Prof
Category:	Computational;#Analysis;#Literature
Software:	Matlab/Python (essential);Abaqus (desired)
Confidential:	No
Offered to:	
	MSc in Composites

Structural health monitoring techniques offer continuous monitoring of structures with permanently mounted sensors, to detect and characterize damage. This is of particular interest of aerospace structures where SHM can result in reduced maintenance time and cost while increasing its safety.

Fibre optic (FO) sensors are one of the main sensor technologies that are being used for SHM. There are various types of FO sensors, based on their sensing technology. This

project different FO sensors will be critically assessed. Existing sensor data will be used for comparing the response of different systems together with a thorough literature review and a critical assessment. The project will also include some modelling (e.g. reflected spectrum of the FO sensors) and hence knowledge of programming such as Matlab will be required.

Sherwin, Spencer Prof

Project no:	IC-MSCSS01
Project title:	Software Engineering in open source CFD package Nektar++
Supervisor:	Sherwin, Spencer Prof
Category:	Computational;#Numerical; Survey
Software:	GitLab Nektar++ Paraview/VisIt
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;#
	MSc in Advanced Computational Methods

Software engineering is an integral part of any computational package. For example, to improve the robustness and flexibility of the Nektar++ (www.nektar.info) flow solver and to help provide an easier way to understand the fundamentals of the spectral/hp element methods we have been developing a python interface. We also maintain a Continual Integration environments within gitlab (gitlab.nektar.info) which helps us ensure that previous developments are robust with respect to new developments. However we do not currently have a performance monitoring framework. This project is therefore focussed on the software engineering aspects of the code development and can be tailored depending on mutual discussion on your interests and the needs of the project.

Project no:	IC-MSCSS02
Project title:	Computational fluid dynamics using spectral/hp element methods
Supervisor:	Sherwin, Spencer Prof
Category:	Computational
Software:	Nektar++ VisIt or Paraview
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Projects are available in the development and application of high order finite element methods known as spectral/hp element discretisations. These project will be based around the library, Nektar++, details of which can be found under (www.nektar.info). The primary focus of the projects will be the development and/or application of these techniques to incompressible and compressible flow problems similar to those shown under

www.nektar.info. Although there are no specific prerequisites for the projects priority will be given to students with computing and coding skills.

Steiros, Kostas Dr

Project no:	IC-MSCKS01
Project title:	Turbulence fluctuations in a wind farm
Supervisor:	Dr Kostas Steiros
Category:	Theoretical;#Analysis;#Literature
Software:	starccm+
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

According to a recent Science review article, a grand challenge in the science of wind energy remains the accurate modelling of the turbulent fluctuations that wind turbines experience in farms. The difficulty lies on the fact that while turbulent fluctuations are currently modelled using the well-known -5/3 law, the latter is only valid in equilibrium regimes – a condition which is far from fulfilled in the ever-changing atmospheric conditions that a farm experiences. Recent investigations at Imperial College have shown that such out-of-equilibrium dynamics can be modelled using an appropriate correction to the -5/3 law. A natural question then is, do wind farm fluctuations abide to this correction? To ascertain if this is true, the students will analyse two datasets: one from a wind tunnel experiment of a wind farm conducted at Johns Hopkins University, and one from a wind farm high-fidelity simulation, using the code Xcompact 3d of ICL (this project is appropriate for two students, one for each dataset). The data include velocity, force, and power. The goal will be to find an optimal way to model the wind farm turbulent gusts, and compare experimental and numerical results.

Project no:	IC-MSCKS02/E
Project title:	Frequency of vortex shedding
Supervisor:	Kostas Steiros
Category:	Experimental;#Computational;#Theoretical;#Analysis;#Numerical
Software:	starccm+
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

When a stream of fluid encounters a bluff body, vortices of opposite sign emanate sequentially from the edges of the body, forming a vortex street. This is the well-known phenomenon of vortex shedding, which has remained the focus of intense investigations for more than a century. An interesting characteristic of vortex shedding is that the emanating vortices appear at a particular frequency, which is unique for each body shape (e.g. cylinder, flat plate, wedge). What exactly is the cause of this frequency locking is not known, however

it is postulated that it has something to do with the vortex strength that each body shape can create. For instance, cylinders create weaker vortices than flat plates. Assuming that this rationale holds, then if we could attenuate the vortices of the plate, we would be able to recover the frequency of the cylinder. In this project, the students will test this conjecture. By positioning an interference element in the wake of a flat plate, the vortices will be attenuated. The resulting frequencies will be documented, and will be compared with those of other shapes. This project is available for two students. One will conduct the experimental investigation (Particle Image Velocimetry in the Aeronautics flume) and one will simulate the flow using the package Star-CCM+.

Project no:	IC-MSCKS03
Project title:	Aerodynamics of confinement
Supervisor:	Kostas Steiros
Category:	#Computational;#Theoretical;#Analysis;#Numerical; #Literature
Software:	starccm+
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

External aerodynamics deal with bodies in unconfined media, e.g. wind turbines, aircraft wings and automobiles. However, when these flows are tested in wind tunnels or simulated in numerical domains, the flow quantities are affected by the unwanted effect of the tunnel/numerical domain walls. To counter that, all measurements have to be corrected using blockage correction models. For the case of mean velocities and drag, current blockage models perform quite well. However, no model exists for the correction of the turbulence quantities (e.g. turbulence intensity, Reynolds stresses). To address that issue, this project proposes a combined experimental/numerical investigation, conducted by two students. The first student will process PIV and drag measurements from an existing dataset of bluff bodies, measured in various blockage levels in a water channel. The second student will simulate these configurations using the package Star-CCM+. The end goal is the development of a novel low order blockage correction model.

Tagarielli, Vito Dr

Project no:	IC-MSCVT01/E
Project title:	High strain rate testing of engineering solids – Project 1
Supervisor:	Vito Tagarielli
Category:	Experimental;#Experimental: Structures;#Design;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

The objective is to assemble a split Hopkinson pressure bar (SHPB), to commission and calibrate this apparatus, and to conduct tests with it. The project is open to up to 4 students;

these will initially work as a group until the equipment is completely operational; then, each student will use the SHPB to conduct high-rate compression tests on different types of high-performance materials, to be determined later in the project (we envisage these will be composites, foams or polymers).

It will be necessary to complete all the assembly/commissioning work quickly to ensure a smooth second part of the project.

The activities will involve using different types of equipment competently and safely. Regular attendance of the college will be required throughout the project. No remote working is possible.

The project suits students with sound understanding of applied mechanics, capable of liaising effectively with colleagues and technicians, and willing to put some hard work in the activities, which could result in publishable research. The student will pick up a basic technique in experimental solid mechanics and learn the principles of high speed photography and data acquisition, along with the response of the particular material/structure tested.

Project no:	IC-MSCVT02/E
Project title:	High strain rate testing of engineering solids – Project 2
Supervisor:	Vito Tagarielli
Category:	Experimental;#Experimental: Structures;#Design;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

See project description above.

Project no:	IC-MSCVT03/E
Project title:	High strain rate testing of engineering solids – Project 3
Supervisor:	Vito Tagarielli
Category:	Experimental;#Experimental: Structures;#Design;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

See project description above.

Project no:	IC-MSCVT04/E
Project title:	High strain rate testing of engineering solids – Project 4
Supervisor:	Vito Tagarielli
Category:	Experimental;#Experimental: Structures;#Design;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Composites

See project description above.

Wynn, Andrew Dr

Project no:	IC-MSCAW01
Project title:	Control, stability and data-driven modelling of flexible structures and fluid structure interactions
Supervisor:	Dr Andrew Wynn
Category:	Computational;#Theoretical;#Analysis;#Numerical
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This project will study the modelling, analysis and control of partial differential equations which underpin the dynamics of very flexible aircraft. Next generation hybrid electric or fully electric aircraft, will employ very high aspect ratio wings which will necessarily deform significantly during flight. As a result, typical methods of linear structural analysis, wake modelling (e.g. linear UVLM), and linear control design (e.g. LQR) are unlikely to be sufficient to guarantee flight integrity: relying on such approaches has led to some famous failures such as NASA's Helios mishap (https://www.nasa.gov/pdf/64317main_helios.pdf). These projects will take a nonlinear modelling approach, and instead employ Geometrically Exact Beam Theory (GEBT) to model very flexible aircraft. GEBT models can describe arbitrarily large nonlinear deformations, and contain classical Euler-Bernoulli and Timoshenko models. However, in contrast to classical model, the underlying mathematical properties of the nonlinear PDEs underpinning GEBT are significantly less developed. For modelling of fluid-structure interactions we will use medium fidelity UVLM-based aerodynamic models which can be easily coupled with GEBT models.

A number of open research questions in this area exist, one of which will be chosen for the project:

1. Deriving verifiable conditions for stability and regularity (i.e. to avoid buckling) of nonlinear flexible beam models. Due to the underlying nonlinearity of the PDE, a possible approach is to attempt to construct Lyapunov functions using Sum-of-Squares programming.
2. Stability and stabilizability of flexible aeroelastic systems. This project will seek to couple flexible beam models with aerodynamic models (e.g. reduced-order UVLM) and investigate the stability, flutter, and robustness of such systems. Again, a Lyapunov-based approach is expected to be necessary.

3. Data-driven modelling of fluid structure interactions. Even a medium fidelity aerodynamic model such as UVLM requires a large number of states ($\sim 10^5$) to accurately capture the necessary wake dynamics. We will aim to use a data-driven approach to learn minimal nonlinear reduced-order models which capture the key physical dynamics. If successful, having such models will open the door to exploiting state-of-the-art tools from nonlinear control.

4. Model-predictive control (MPC) of flexible structures. MPC is an optimization-based control theoretic methodology that can truly exploit nonlinear effects. This project will seek to develop novel (and efficient) MPC approaches to control flexible structures. It will be advantageous if students selecting this project have studied aeroelasticity (certainly for 2., above) and control systems courses. Ability to program in Matlab is necessary.

Project no:	IC-MSCAW02
Project title:	Understanding Internally Heated Convection using Computational Optimization
Supervisor:	Dr Andrew Wynn
Category:	Computational;#Theoretical;#Analysis
Software:	Matlab
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

A fundamental challenge in fluid mechanics is to predict certain time-average properties of a given flow as a function of the parameters, for instance the average drag coefficient as a function of the Reynolds number. In regimes where experimental data is scarce or fully-resolved numerical simulations are too expensive, one way to make progress is to look for bounds on the properties of interest, rather than exact values.

Bounds can be derived analytically from the Navier-Stokes equations using the “background field method” of Doering & Constantin. Applications of this method require sophisticated mathematical analysis and, often, the bounds one finds have limited predictive power because they are not quantitatively sharp. Recently, however, it has been shown that the background field method can be automated using numerical optimization, with significant improvements on the bounds that can be proven by hand.

Numerical optimization also enables one to handle a powerful extension the background method, which has so far been beyond the reach of analytical work. The aim of this project is to further develop optimization-based methods for bounding the properties of fluid flows with specific application to computing the Heat Transfer in Internally-Heated Convective Flows. The behaviour of these flows is still poorly understood despite being key to many geophysical, astrophysical, and industrial process (see the paper "Convection driven by internal heating", Goluskin & Spiegel, Physics Letters A 377(1-2):83-92, <https://doi.org/10.1016/j.physleta.2012.10.037>). As part of the project, students will derive mathematically an optimization problem for the bounds being sought and solve it numerically using open-source MATLAB software for numerical optimization. The balance between the analytical and numerical aspects of the project is flexible, but students should have a strong mathematical background (or an interest in mathematics) and be proficient with MATLAB.

Project no:	IC-MSCAW03
Project title:	Efficient wake models for wind farms: global yaw steering strategies via polynomial optimization
Supervisor:	Dr Andrew Wynn
Category:	Computational;#Theoretical;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

The power output of a wind farm is highly dependent upon interactions of the wakes generated individual turbines. In particular, a downstream turbine which lies directly in the wake of an upstream machine may produce up to 45% less power, due to the lower inflow velocity that it experiences. To attempt to mitigate this effect "wake steering" can be used, which involves intentionally yawing each turbine in a farm to reduce the impact of low-momentum wakes on downstream machines. However, since yawing reduces the power of the upstream machine, there is necessary a trade-off, which leads to an optimization problem to maximize wind farm power. Due to the complexity of wind farm wakes, such optimization problems are challenging to solve and may possess many local maxima. This project will take a new approach by seeking to build simpler models for turbine wakes from existing complex models (e.g. the recent model doi:10.1017/jfm.2016.595). Models will be constructed which are specifically tailored to creating easy-to-solve convex optimization problems (e.g. those with polynomial cost functions and constraints), in order to find globally optimal wake steering strategies. Further extensions of this project will consider wake unsteadiness and unsteady inflow conditions to wind arrays in an optimization context.

Project no:	IC-MSCAW04
Project title:	Adaptive data-driven flow field estimation for wind farms
Supervisor:	Dr Andrew Wynn
Category:	Computational;#Theoretical;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Wind farms are subject to unsteady, turbulent atmospheric inflow conditions. However, it is standard practice to optimize the design and operational settings of a wind array by only considering very high-level information about the flow conditions such as the expected mean wind direction. Consequently, if more detailed information about the flow field within a wind farm can be inferred from available sensor readings, then smarter decisions can be made at the level of individual wind turbine control to improve performance (e.g. maximize power output, or minimize fatigue). A barrier to flow-field estimation is that sensor information is sparse, typically limited to sensors located on the turbines themselves, meaning that no information about the flow between turbines is directly measured. This project aims to overcome this barrier. The aim is to use state-of-the-art, time-varying, PDE models for turbine wakes (see <https://onlinelibrary.wiley.com/doi/10.1002/we.2093>) coupled with

adaptive estimation techniques (e.g. Moving Horizon Estimation) which solve online nonlinear optimization problems for both parametric identification and state (flow field) reconstruction.

Project no:	IC-MSCAW05/E
Project title:	Real-time control of wind turbines for farm power maximization
Supervisor:	Dr Andrew Wynn
Category:	Experimental
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

This research is related to the project “efficient wake models for wind farms: global yaw steering strategies via polynomial optimization”. The main distinction being that this project will consider control strategies which respond in real time to unsteady flow conditions, while the former project seeks time-independent wake control via polynomial optimization. In this project, wind farm wakes will be approximated using the PDE models of (<https://onlinelibrary.wiley.com/doi/10.1002/we.2093>). Real-time control strategies, such as Model Predictive Control, will be employed to solve problems such as farm power maximization, power output tracking, and fatigue reduction. Since MPC requires solution of online optimization problems, efficient adjoint-based optimization strategies will also be developed as part of this project, exploiting the advective structure of the underlying PDE.

Project no:	IC-MSCAW06/E
Project title:	Optimization-based analysis of polymer induced drag reduction in fluid flows
Supervisor:	Dr Andrew Wynn
Category:	Experimental
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

It is well known that the addition of polymers to fluid flows can significantly reduce drag, achieving the state known as maximal drag reduction. Although this phenomenon has been 99 observed experimentally, much less is known about why polymer-induced drag reduction occurs. This project will use computational optimization to attempt to understand this property from a mathematical perspective. This will involve studying partial differential equation (PDE) models for polymer flows, such as Oldroyd-B and FENE-P models, which are adaptations of the Navier-Stokes equations. Interested students should read: [C. R. Doering, B. Eckhardt, and J. Schumacher. Failure of energy stability in Oldroyd-B fluids at arbitrarily low Reynolds numbers. *J. Non-Newtonian Fluid Mech.*, 135(2–3):92–96, 2006]. This project is appropriate for students who enjoy mathematical aspects of fluid dynamics, PDEs and optimization. It requires a strong level of mathematical ability.

Project no:	IC-MSCAW07
Project title:	Data-driven modelling, estimation, and control of turbulent fluids (Project 1)
Supervisor:	Dr Andrew Wynn
Category:	Computational;#Theoretical;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

Creating a low-dimensional model of a fluid flow is often the first step towards the design of a flow control strategy to improve performance (e.g. reduce drag). Due to the complexity of the Navier-Stokes equations, it is often desirable to extract flow models directly from large ensembles of fluid flow data (e.g. from numerical simulations or experiments). In this data-driven approach, model construction is achieved using computational optimization. Many strategies are available, ranging from classical linear programming to deep neural networks. The main open research question, which will be investigated in these FYPs, is to find the appropriate balance between the complexity of the fitted model and its off-training-set predictive capability. The chaotic dynamics of turbulent flows have, to-date, prevented resolution of this question. In these projects, data from bluff body flows and turbulent boundary layers (among other flows) will be considered to resolve it. Experience of Matlab is necessary. Two projects are available on this subject.

Project no:	IC-MSCAW08
Project title:	Data-driven modelling, estimation, and control of turbulent fluids (Project 2)
Supervisor:	Dr Andrew Wynn
Category:	Computational;#Theoretical;#Analysis
Software:	
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods

See project description above.

Yang, Daqing Dr

Project no:	IC-MSCDY01
Project title:	CFD analysis for flying underwater glider
Supervisor:	Dr Daqing Yang
Category:	Design;#Analysis;#Numerical
Software:	CATIA, Solidworks, Star CCM+, Fluent
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

This project will investigate the possibility of converting an underwater glider into a flying aircraft with emphasis on aerodynamic and hydrodynamic analysis. This amphibious vehicle will be able to deploy as an underwater glider, take-off from underwater, fly and loiter in atmosphere with a certain duration, land back to water and resume its underwater voyage. Student is expected to use CFD software to analyze both underwater and flying aero-hydrodynamic properties of the glider. Key focus of the study is the multi-phase flow interaction during the take-off phase.

Project no:	IC-MSCDY02
Project title:	Flying underwater glider conceptual and preliminary design
Supervisor:	Dr Daqing Yang
Category:	Design;#Analysis
Software:	XFLR5, NeoCASS, CATIA, MATLAB, Flight Gear
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

Underwater glider is a type of autonomous vehicle that employs variable-buoyancy propulsion with a pair of hydrofoils. It is widely used in ocean research organizations. This project will investigate the possibility of converting an underwater glider into a flying aircraft. This amphibious vehicle will be able to deploy as an underwater glider, take-off from underwater, fly and loiter in atmosphere with a certain duration, land back to water and resume its underwater voyage. Student is expected to design both underwater and flying aero-hydrodynamic, structure, propulsion and control aspects of the vehicle. A flight simulation is also required for demonstration of this novel concept.

Project no:	IC-MSCDY03/E
Project title:	e-VTOL aircraft design for urban air mobility
Supervisor:	Dr Daqing Yang
Category:	Experimental: Manufacturing intensive;#Design;#Analysis
Software:	XFLR5, NeoCASS, CATIA, MATLAB, Flight Gear, PX4 autopilot, Mission Planner
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

Combination of vertical take-off and landing aircraft with electric propulsion has demonstrated huge potential in future urban air mobility applications, this project will review current design proposals, endeavor a novel design concept and test its feasibility with off-the-shelf drone components. Simulation is required as physical presentation of this vehicle in operation. Flight and wind tunnel experiments will be needed to verify the design outcome.

Project no:	IC-MSCDY04/E
Project title:	Distributed electric-powered foot launch hang glider design
Supervisor:	Dr Daqing Yang
Category:	Experimental: Manufacturing intensive; #Computational;#Design;#Analysis;#Literature
Software:	CATIA, MATLAB, Star CCM+, Flight Gear, PX4 autopilot, Mission Planner
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;
	MSc in Advanced Computational Methods
	MSc in Composites

This project focus on physical development of a scaled electrically powered foot launch hang glider design. The development will utilize off-the-shelf model aircraft and drone components. Simulation is required as physical presentation of this vehicle in operation. Flight or wind tunnel experiments will be needed to verify the design outcome.

Project no:	IC-AAEDY05
Project title:	Preliminary design of integrated hydrogen fuel tank for blended wing body aircraft
Supervisor:	Dr Daqing Yang
Category:	Computational;#Design;#Analysis;#Literature
Software:	CATIA, Abacus, Fusion 360
Confidential:	No
Offered to:	MSc in Advanced Aeronautical Engineering;

This project will investigate the feasibility of an integrated hydrogen fuel tank design for a blended wing body aircraft. The design will combine the tube-shaped hydrogen tank with

load bearing structures to maximize structural efficiency and reduce airframe weight penalties due to adoption of hydrogen fuel. Student will be given a BWB aircraft conceptual design and focus on the design and optimization of the airframe and integrated fuel tank.